# PMP® Demo Exam 08

## Based on the *PMBOK® Guide – Eighth Edition* and the PMP® Examination Content Outline (July 2026)

| Item | Real exam | This demo exam |
|---|---|---|
| Questions | 180 (170 scored + 10 unscored pretest) | 180 (all scored) |
| Time | 240 minutes | 240 minutes |
| Breaks | Two optional 10-minute breaks | After Q60 and after Q120 |
| Domains | People 33% · Process 41% · Business Environment 26% | Same weighting |
| Approaches | ~40% predictive, ~60% agile/hybrid | Same mix |

**How to take this exam:** set a timer for 240 minutes, sit somewhere quiet, and answer every question without notes. Write your answers on paper (question number + letter). There is no penalty for guessing, so never leave a question blank. Pace yourself at about **80 seconds per question** — you should be near Q60 at 80 minutes and Q120 at 160 minutes. Only after finishing, check the Answer Key and read the explanations, then complete the score analysis.

*Question types:* most questions have one best answer. Multiple-response questions say how many options to choose. Matching questions ask you to pair items with options. Fill-in-the-blank questions need a number. Case-study sets share one scenario across several questions.

*On screen:* the same exam runs in the PMP Exam Simulator (`04_PMP_Exam_Simulator.html`), which times it, scores it by domain and task, and then gives you a PDF with the correct answer and the explanation for every question you missed.

## Section 1 — Questions 1 to 60

**Question 1.** Several key suppliers close for two weeks during a national holiday in their country. Where should this be reflected?

- **A.** Nowhere, since it is outside the project
- **B.** In the calendars used for scheduling
- **C.** In the issue log, once it happens
- **D.** In the stakeholder register only

**Question 2.** Citizens complain that the new parking sensors are issuing incorrect fines. What should the project manager do?

- **A.** Ask citizens to appeal each fine individually through the city's normal appeals process
- **B.** Wait for the vendor's next scheduled software update to fix the sensor problem
- **C.** Defend the system publicly, since the sensors passed all of their tests before going live
- **D.** Investigate quickly, fix or suspend the faulty function and put things right

**Question 3.** Early-access players expect weekly updates, but the team can sustain only monthly updates of good quality. What should the product owner do?

- **A.** Communicate a realistic update cadence and roadmap, and keep to it
- **B.** Release weekly updates, even if their quality is lower than usual
- **C.** Stop communicating about updates to avoid creating expectations
- **D.** Promise weekly updates, since players will leave if updates are slow

**Question 4.** The game's direction is dominated by the chief executive's personal opinions, even when player tests show something different. What should the product owner do?

- **A.** Stop inviting the chief executive to reviews so that the team can follow the data alone
- **B.** Run more tests until the results agree with the chief executive's personal opinions
- **C.** Bring player data and tests into decisions, while keeping the CEO engaged
- **D.** Follow the chief executive's opinions, since the chief executive is the most senior person

**Question 5.** New buildings must meet an energy performance standard before they can be certified for use. How should the project ensure compliance?

- **A.** Check compliance once the building is complete, and make any changes that are needed then
- **B.** Apply for an exemption, since meeting the standard would add cost to the project
- **C.** Build it into the design requirements and verify it at reviews and before handover
- **D.** Ask the contractor to confirm in writing that the building will meet the energy standard

**Question 6.** Several subcontractors are in financial difficulty because of an economic downturn. What should the project manager do?

- **A.** Ignore the situation, since the subcontractors are responsible for their own finances
- **B.** Monitor their finances, identify alternatives and consider protections such as bonds
- **C.** Pay the subcontractors in advance to help them through their difficulties
- **D.** Terminate all of the subcontracts now, before any of the subcontractors fail

**Question 7.** Two months before launch, designers keep proposing new features. The team agreed a content lock at this point. What should the product owner do?

- **A.** Let the designers decide which of their new features can still be added before the lock
- **B.** Extend the launch date so that the best new features can be included
- **C.** Accept the new features, since good ideas should never be postponed
- **D.** Apply the content lock and add the new ideas to the post-launch backlog

**Question 8.** Match each type of dependency with its description.

*(Matching: pair every numbered item with one lettered option.)*

| Item | Option |
|---|---|
| **1.** Mandatory | **A.** Based on best practice or the team's preference |
| **2.** Discretionary | **B.** Involves activities within the project's control |
| **3.** External | **C.** Involves activities outside the project |
| **4.** Internal | **D.** Required by the contract or the nature of the work |

**Question 9.** The developer wants to change the façade material after construction has started. What formal instrument is used with the contractor?

- **A.** A new tender for the whole building project
- **B.** An email asking the contractor to start the work
- **C.** A variation order agreed under the contract
- **D.** A verbal instruction from the developer on site

**Question 10.** Messages between the city and three vendors get lost, and the vendors contact different city officials. What should the communications management plan define?

- **A.** A weekly meeting with every city official and all vendors
- **B.** A rule that vendors may contact any city official they like
- **C.** A ban on direct communication between the city and vendors
- **D.** Single points of contact and agreed communication protocols

**Question 11.** Which two activities prevent defects rather than detect them? (Choose two.)

- **A.** Analysing customer complaints after delivery
- **B.** Training staff in the procedures before they start work
- **C.** Designing the process so that it consistently produces quality
- **D.** Inspecting the final product before it is shipped
- **E.** Walking through each apartment to list defects

**Question 12.** The same supplier problem is affecting four projects in the programme. What should the project manager do?

- **A.** Solve it within this project only, since the other projects have their own managers
- **B.** Change supplier for this project immediately, without telling the other project managers
- **C.** Wait to see how the other projects solve it and then copy their approach
- **D.** Escalate it to programme management so that it can be handled once for all projects

### Case Study — Harbourline smart district

Harbourline City is turning its old port area into a smart district, with sensors for parking, air quality and pedestrian flows, a citizen app and 5G small cells. Civil works follow a predictive plan; the data platform and the app are built by agile teams.

- The data protection authority has issued guidance: **video analytics in public spaces** must be limited to anonymised counting, and a **data protection impact assessment** (DPIA) is required.
- A **ransomware attack** on a neighbouring city's systems has raised concerns about the district's cybersecurity.
- **Parking enforcement officers** will move from street patrols to monitoring dashboards and responding to alerts. Many are anxious about the change.
- The national government has announced new **5G security rules** that restrict equipment from certain vendors, including the vendor the project selected.

*Questions 13–16 refer to this case.*

**Question 13.** *(Case study: Harbourline smart district)* How should the project respond to the guidance on video analytics?

- **A.** Rely on the vendor's statement that its analytics software already complies with the guidance
- **B.** Collect full video now and anonymise it later, once the system has proved that it is useful
- **C.** Complete the DPIA early and design the analytics for anonymous counting only
- **D.** Collect the video for research purposes only, since research is exempt from the guidance

**Question 14.** *(Case study: Harbourline smart district)* Which response best addresses the cybersecurity risk?

- **A.** Transfer the risk entirely to the vendors through clauses in their contracts with the city
- **B.** Build security into the design and testing, and prepare a tested incident response plan
- **C.** Avoid the risk by keeping every sensor and system disconnected from any network
- **D.** Accept the risk, since the attack on the neighbouring city used different systems

**Question 15.** *(Case study: Harbourline smart district)* Which message is most likely to build the enforcement officers' desire to adopt the new way of working?

- **A.** That the decision has been made by the council and cannot be discussed with the officers
- **B.** That officers who do not adapt quickly may be moved to other roles in the city
- **C.** How the change benefits them, such as safer work and less patrolling in bad weather
- **D.** That the technology is very advanced and has been used successfully in other countries

**Question 16.** *(Case study: Harbourline smart district)* How should the project respond to the new 5G security rules?

- **A.** Cancel the 5G part of the project without assessing any alternatives
- **B.** Assess the legal impact, engage the vendor, identify alternatives and use change control
- **C.** Continue with the selected vendor, since the contract was signed before the rules
- **D.** Install the equipment quickly, before the new security rules come into force nationally

**Question 17.** The contractor encounters unexpected rock and submits a claim under the contract's clause on differing site conditions. What should the project manager do?

- **A.** Assess the claim against the contract and the evidence, and negotiate a fair settlement
- **B.** Reject the claim, since the contractor should have expected rock on the site
- **C.** Accept the claim in full immediately, so that any kind of dispute with the contractor is avoided
- **D.** Ask the contractor to continue and discuss the claim once the project has ended

**Question 18.** Investors want monthly financial reports, while future tenants want to give input on the design. What should the stakeholder engagement plan do?

- **A.** Engage only the tenants, since they will use the building every day
- **B.** Tailor the methods and frequency of engagement to each group
- **C.** Give every group the same monthly financial report for consistency
- **D.** Engage only the investors, since they fund the development project

**Question 19.** In a hybrid smart city project, who accepts the agile increments and who accepts the contractual milestones?

- **A.** The scrum master accepts the increments, and the vendor accepts the contractual milestones
- **B.** The product owner accepts the increments; the city formally accepts the milestones
- **C.** The city accepts both, at the end of the project only
- **D.** The developers accept both, since they know the work best

**Question 20.** The general contractor and the architect blame each other for a design error that requires rework. The dispute is starting to delay the work. What should the project manager do?

- **A.** Wait until the dispute has been settled before any rework is started on the building
- **B.** Agree the fix now and settle responsibility separately under the contract
- **C.** Ask the architect to pay for the rework, since the design contained the error
- **D.** Ask the general contractor to pay for the rework, since it built the faulty element

**Question 21.** Which two characteristics make a product vision effective? (Choose two.)

- **A.** It is short and memorable enough to guide everyday decisions
- **B.** It lists every feature that the product will eventually contain
- **C.** It includes the detailed schedule and the budget for development
- **D.** It describes the target customers and the value they will get
- **E.** It is written by the legal department to avoid any commitments

**Question 22.** A new steering committee keeps debating what it is allowed to decide. What should be agreed?

- **A.** A longer meeting schedule, so that the committee has more time for its debates
- **B.** A decision to dissolve the committee, since the project manager can decide alone
- **C.** Terms of reference: purpose, membership, decision rights and meeting rules
- **D.** A rule that the committee chair decides everything, so that meetings are shorter

**Question 23.** Workstream leads choose their own red, amber and green status colours, so the statuses are inconsistent. What should the project manager do?

- **A.** Define objective thresholds for each colour on each measure

- **B.** Ask the leads to report green unless there is a crisis that needs escalation
- **C.** Choose the colours personally for every workstream
- **D.** Remove the colours and report only in written comments

**Question 24.** Drug development follows regulatory stage gates, while the data science team works in sprints. How should the plans be integrated?

- **A.** Align the sprint outputs with the evidence needed for each stage gate
- **B.** Keep the sprint plans separate from the stage gates to avoid confusion
- **C.** Make the data science team follow the stage gates instead of sprints
- **D.** Ask the regulator to accept sprint reviews instead of the stage gates

**Question 25.** Work often stalls because nobody knows when an item may move from "review" to "test" on the team's board. What should the team do?

- **A.** Add more columns to the board so that the work is described in detail
- **B.** Make explicit policies for each column, such as entry and exit criteria
- **C.** Remove the review column so that items move straight to testing without waiting
- **D.** Ask the product owner to decide each time whether an item can move

**Question 26.** A self-managing team wants a decision approach in which a proposal is adopted unless someone raises a reasoned objection. What is this called?

- **A.** Autocratic decision-making
- **B.** Consent-based decision-making
- **C.** Unanimous agreement
- **D.** Majority voting

**Question 27.** Before committing to full production, the publisher asks for a "vertical slice" of the game. What is its purpose?

- **A.** To complete the most technically difficult engine work before the design is started
- **B.** To produce marketing screenshots of every level of the game before production begins
- **C.** To prove the game's quality, fun and production pipeline on a small but complete section
- **D.** To estimate the cost of the whole game by building its first level to a rough draft standard

**Question 28.** The planned and actual cumulative cost and progress curves show the actual progress curve below the plan and moving further away each month. What does this indicate?

- **A.** The project is ahead of schedule, since less work has been needed than was planned
- **B.** The project is falling further behind; the causes should be investigated and acted on
- **C.** The plan was too optimistic, so the baseline should be adjusted to match the actuals
- **D.** Nothing, since the actual progress curve always lags behind the plan during construction

**Question 29.** Which practice best reduces expectation gaps when apartments are sold before they are built?

- **A.** Attaching detailed specifications to the sales contracts and showing a model apartment
- **B.** Using attractive marketing images that show the apartments at their very best
- **C.** Letting sales agents describe the finishes verbally to each of the buyers
- **D.** Keeping the specifications vague, so that the developer can stay flexible during construction

**Question 30.** What shape does a cumulative, time-phased cost baseline usually have?

- **A.** A flat horizontal line
- **B.** A bell curve
- **C.** An S-curve
- **D.** A straight vertical line

**Question 31.** In principled negotiation, what should the negotiators focus on?

- **A.** Their opening positions
- **B.** Interests rather than positions
- **C.** Winning as much as possible
- **D.** The other side's weaknesses

**Question 32.** A city wants to install smart streetlights in every district, but the technology is new to it. How should the rollout be planned?

- **A.** Install the streetlights in every district at the same time to finish quickly
- **B.** Pilot in one district with clear criteria, then scale based on results
- **C.** Let each district plan and install its own streetlights independently
- **D.** Wait until another city has completed a full rollout before starting the work

**Question 33.** Tenants of an existing building must be relocated while it is redeveloped. How should the project engage them?

- **A.** Through the building's owner only, since the tenants have no contract with the project
- **B.** Early, with clear information, a contact person and practical relocation support
- **C.** Shortly before demolition, so that the tenants have less time to object to the plan
- **D.** Through a legal notice, since the tenants' leases allow the owner to redevelop

**Question 34.** A new project manager introduces strict rules on core working hours without consulting the self-organising team. Morale drops. What should the project manager have done?

- **A.** Applied the rules only to the newest members of the team
- **B.** Facilitated the team in defining its own working agreements
- **C.** Introduced the rules gradually over several sprints instead
- **D.** Asked human resources to announce the rules on his behalf

**Question 35.** Why can the SPI be misleading near the end of a project that is running late?

- **A.** Because SPI measures cost efficiency, not schedule performance, in the final phase
- **B.** Because SPI moves towards 1.0 as work is completed, even if the project is late
- **C.** Because SPI is calculated from actual costs, which are not final until the very end
- **D.** Because SPI can only be calculated for projects that are ahead of their schedule

**Question 36.** How should payments to a co-development studio be structured to encourage working results?

- **A.** Pay the full amount in advance to build trust with the studio
- **B.** Pay only at the end, once the whole game has been released
- **C.** Tie them to accepted playable builds at agreed milestones
- **D.** Pay for hours worked, whatever the studio actually delivers

**Question 37.** Which closure activity reduces ongoing costs after the project ends?

- **A.** Extending the team members' contracts for another quarter as a precaution
- **B.** Renewing all of the project's software subscriptions
- **C.** Cancelling project-specific contracts, licences and facilities
- **D.** Keeping the project office open in case it is needed again

**Question 38.** Rising interest rates could make the development's financing much more expensive. What should the project manager do?

- **A.** Ignore the risk, since interest rates are outside the project's control
- **B.** Wait until rates have risen before discussing the problem with finance
- **C.** Speed up construction by cutting some quality checks, so that less interest is paid overall
- **D.** Work with finance on responses, such as fixing the interest rate or hedging

**Question 39.** A major platform announces that it will change its payment rules, which affects the game's in-game purchases. What should the product owner do?

- **A.** Assess the impact on the business model and adapt the backlog and roadmap
- **B.** Continue as planned, since the new rules will not take effect until next year at the earliest
- **C.** Ask players to make purchases outside the platform to avoid the fees
- **D.** Remove the game from the platform immediately to avoid the new rules

**Question 40.** The team wants everyone to see immediately when the automated build breaks. What should it use?

- **A.** A note in the sprint review presentation
- **B.** A report sent to the product owner monthly
- **C.** A visible dashboard showing the build status
- **D.** A weekly email summarising build failures

**Question 41.** After the game launches, players flood the forums with complaints about server outages. What should the studio do?

- **A.** Blame the hosting provider publicly so that players do not blame the studio
- **B.** Delete the most negative posts so that new players are not discouraged
- **C.** Acknowledge the problem quickly and share regular, honest updates on the fixes
- **D.** Wait until the servers are stable before saying anything to the players

**Question 42.** At a public meeting about new traffic cameras, a citizens' group and the city's transport department start arguing loudly. What should the project manager do?

- **A.** End the meeting immediately and continue the discussion with each group separately by email
- **B.** Calm the discussion, acknowledge the concerns and refocus on the shared goal of road safety
- **C.** Let both sides argue until they run out of energy, since this will reduce tension in future
- **D.** Support the transport department, since it is the project's sponsor and approved the cameras

**Question 43.** The sensors passed all laboratory tests but fail in heavy rain. What should be changed?

- **A.** Add field tests in realistic conditions to the acceptance criteria
- **B.** Accept the failures, since the sensors passed all of the laboratory tests
- **C.** Install the sensors only in streets that are sheltered from rain

- **D.** Ask the vendor for a discount on the sensors that fail in the rain

**Question 44.** After a merger, workers worried about keeping their jobs show little interest in the project's recognition scheme. Which theory best explains this?

- **A.** Tuckman's ladder of development
- **B.** McGregor's Theory X and Theory Y
- **C.** Vroom's expectancy theory
- **D.** Maslow's hierarchy of needs

**Question 45.** The city now requires its suppliers to provide a list of all software components used in their products. How should the team meet this requirement?

- **A.** Remove all third-party components so that no list is needed at all
- **B.** Ask the city to accept a general description of the main technologies that the product uses
- **C.** Write the list by hand once, at the end of the contract, from memory
- **D.** Generate and maintain a software bill of materials as part of the build pipeline

**Question 46.** Where are the rules for cost control thresholds, units of measure and the level of precision defined?

- **A.** In the project charter
- **B.** In the cost management plan
- **C.** In the stakeholder register
- **D.** In the issue log

**Question 47.** A biotech company wants patients' perspectives on the design of a clinical trial. How should it engage them?

- **A.** Through the regulator, which represents patients' interests in trials
- **B.** Through the doctors only, since patients cannot judge trial designs
- **C.** Through patient advocacy groups, for example in an advisory board
- **D.** Through the trial's marketing materials once recruitment has started

**Question 48.** Three vendors are making conflicting technical decisions about the smart district's data platform. What should the project establish?

- **A.** Separate platforms for each vendor, so that their decisions no longer conflict
- **B.** A design authority that approves technical decisions affecting more than one vendor
- **C.** A rule that the largest vendor makes all of the technical decisions for the platform
- **D.** A monthly meeting where the vendors report the decisions they have already made

**Question 49.** The team needs to estimate 120 backlog items quickly. Which technique fits best?

- **A.** Bottom-up estimating of every task in each item
- **B.** Detailed PERT estimates for each of the 120 items
- **C.** Affinity estimating, grouping items by relative size
- **D.** Asking the product owner to estimate all of the items

**Question 50.** Before a game launch, the art lead wants the team to work overtime, while the engineering lead refuses on behalf of his developers. The disagreement is becoming heated. What should the producer do?

- **A.** Ask the studio director to impose a decision on both leads so that the conflict ends quickly
- **B.** Side with the art lead, since overtime before a launch is normal and expected in game studios
- **C.** Facilitate using data and team norms, and take scope options to the product owner
- **D.** Side with the engineering lead, since developers should never be asked to work any overtime

**Question 51.** The electrical and plumbing subcontractors argue about who should work first in each apartment, and progress is slowing. What should the project manager do?

- **A.** Replace one of the subcontractors with a firm that is more flexible
- **B.** Bring both together to agree a sequenced work plan for each floor
- **C.** Let the subcontractor that arrives on site first each day choose
- **D.** Ask both subcontractors to work in the apartments at the same time

**Question 52.** Too many trades are scheduled on the same floor at the same time, causing congestion and safety risks. What should the project manager do?

- **A.** Resequence the trades so that fewer work on each floor at a time
- **B.** Add more supervisors on each floor so that the congestion can be managed
- **C.** Accept the congestion, since it is normal during the fit-out of a building
- **D.** Ask the trades to work faster so that they leave each floor sooner

**Question 53.** Which two of the following are pull communication methods? (Choose two.)

- **A.** An open data portal
- **B.** A weekly email bulletin
- **C.** A conference call
- **D.** A project intranet site
- **E.** A status meeting

**Question 54.** Misunderstandings between the product owner, developers and testers often cause rework. Which practice builds shared understanding before work starts?

- **A.** A test plan written by the testers after development
- **B.** Short "three amigos" conversations about each story
- **C.** A handover meeting at the end of each sprint
- **D.** A detailed specification written by the product owner alone

**Question 55.** Which two characteristics make a risk statement useful? (Choose two.)

- **A.** It is vague, so that it covers as many situations as possible
- **B.** It describes only the effect, without the event or cause
- **C.** It names the person who would be to blame if it occurs
- **D.** It is specific enough to be analysed and responded to
- **E.** It describes the cause, the uncertain event and its effect

**Question 56.** Stakeholders complain that the last two sprint reviews showed "nothing new", because the team worked on infrastructure. What should the product owner do?

- **A.** Show the value of the technical work, for example through performance results
- **B.** Skip the sprint reviews until there are new features to show stakeholders
- **C.** Ask the team to add small cosmetic features to every sprint for the reviews
- **D.** Explain that infrastructure work is not the stakeholders' concern at all

**Question 57.** The project manager wants a capable team lead to take ownership of the façade work package. How should the work be delegated?

- **A.** Define the outcome, the authority, the resources and the check-in points
- **B.** Ask the team lead to report on every task every day for the first month
- **C.** Delegate the tasks but keep all of the decisions with the project manager
- **D.** Hand over the work and avoid any contact until it has been completed

**Question 58.** Short chat messages between remote team members are often read as rude, causing friction. What should the team do?

- **A.** Ignore the friction, since chat messages are always short
- **B.** Ask the team members who seem rude to apologise publicly
- **C.** Agree communication norms, such as tone and when to switch to a call
- **D.** Ban chat messages and use only email for all of the team's written communication

### Case Study — Pixelforge game launch

Pixelforge Studios is developing a strategy game for PC and two game consoles. The launch date is fixed with the console makers and the publisher's marketing campaign. Three Scrum teams build the game in two-week sprints; an external vendor creates the 3D character models under a time-and-materials contract.

- Before launch, each console maker must certify the game. Certification takes **4 weeks**, and most games need **one resubmission**, which adds **2 weeks**.
- In the closed beta, the game crashes **3.2 times per 1,000 sessions**; the console makers require fewer than **1 per 1,000**. Crash reports show that **five crash types** cause most of the crashes.
- **30%** of the character models delivered by the vendor exceed the polygon budget defined in the art specification attached to the contract.
- After launch, the game will be run as a live service, earning revenue from optional expansions.

*Questions 59–62 refer to this case.*

**Question 59.** *(Case study: Pixelforge game launch)* How far ahead of the launch date should the teams plan to submit the game for certification?

- **A.** One week, since the console makers usually certify games quickly
- **B.** Two weeks, since a resubmission takes two weeks
- **C.** Four weeks, since certification takes four weeks
- **D.** At least six weeks, allowing for one resubmission

**Question 60.** *(Case study: Pixelforge game launch)* What should the teams do first about the crash rate?

- **A.** Keep adding planned features and fix the crashes in a patch after the launch date
- **B.** Ask the console makers to relax their crash requirement for this game
- **C.** Add a manual test pass before each release to find crashes earlier
- **D.** Fix the five most frequent crash types, tracking the crash rate on every build

> ☕ **Optional 10-minute break.** In the real exam, once you start a break you cannot return to the questions of the previous section, so review flagged questions before you go.

## Section 2 — Questions 61 to 120

**Question 61.** *(Case study: Pixelforge game launch)* How should the project handle the character models that exceed the polygon budget?

- **A.** Change the art specification to match the models, since the vendor's artists know best
- **B.** Accept the models and ask the studio's own artists to quietly fix them, so that the vendor is not upset
- **C.** Terminate the contract immediately, since the vendor has failed to meet the agreed specification
- **D.** Reject the non-conforming models under the contract specification and address the cause with the vendor

**Question 62.** *(Case study: Pixelforge game launch)* After launch, which measure best shows whether the game keeps delivering value to its players?

- **A.** The number of copies sold in the first week after the launch, across all platforms
- **B.** The number of developers working on the live team
- **C.** Player retention over time, such as the share still playing after 30 days
- **D.** The number of features shipped by the live team each month

**Question 63.** A senior producer is leaving the studio next month. Which activity best preserves her knowledge of the key stakeholder relationships?

- **A.** A written list of her stakeholders' names, roles and email addresses for the successor
- **B.** An exit interview with human resources about why she is leaving the studio
- **C.** A farewell party where she can share some stories with the team
- **D.** A structured handover to her successor, including introductions to key stakeholders

**Question 64.** Replacing streetlights with LEDs saves energy immediately, while the sensor network adds further value later. How should the work be sequenced?

- **A.** Wait until the sensors are ready, so that each street is done once
- **B.** Install the sensor network first, since it is the more innovative part
- **C.** Install the LEDs first, so that the energy savings start early
- **D.** Install both at the same time in each street, whatever the cost

**Question 65.** Who approves changes to the project charter?

- **A.** The change control board's secretary
- **B.** The project manager, who wrote it
- **C.** Any member of the project team
- **D.** The sponsor, or the body that issued it

**Question 66.** What does the sprint backlog contain?

- **A.** Every item that the product might ever need in the future, in priority order
- **B.** The list of defects found by customers since the last release
- **C.** The scope baseline approved by the change control board
- **D.** The sprint goal, the selected items and a plan for delivering them

**Question 67.** Match each leadership style with its description.

*(Matching: pair every numbered item with one lettered option.)*

| Item | Option |
|---|---|
| **1.** Transformational | **A.** Puts the team's needs first and removes obstacles |
| **2.** Transactional | **B.** Inspires with a vision and encourages innovation |
| **3.** Servant | **C.** Leaves decisions largely to the team, with little involvement |
| **4.** Laissez-faire | **D.** Uses rewards and corrective action based on performance |

**Question 68.** Before building a new feature for property managers, the product owner wants to validate that the problem really exists. What should she do first?

- **A.** Ask the developers whether the problem sounds real
- **B.** Add the feature to the roadmap and wait to see what feedback arrives
- **C.** Interview a sample of property managers about the problem
- **D.** Build the feature and measure how often it is used

**Question 69.** Several game teams each build their own tools for the same engine tasks, duplicating effort. What should be considered?

- **A.** A small platform team that provides shared tools to the other teams
- **B.** Banning the use of custom tools in all of the teams
- **C.** Merging all of the game teams into one very large team with a single backlog
- **D.** A rule that each team must build all of its tools from scratch

**Question 70.** The project manager must decide quickly how to handle a vendor problem, but wants to benefit from the team's knowledge. Which decision-making style fits best?

- **A.** Autocratic: decide alone without asking anyone
- **B.** Consensus: wait until everyone fully agrees
- **C.** Consultative: gather the team's input, then decide
- **D.** Delegated: let the vendor decide on its own

**Question 71.** Safety briefings on a building site must reach workers who speak five different languages. What should the project manager do?

- **A.** Give each worker a written copy of the briefing in the local language to sign
- **B.** Ask the workers who speak the local language to pass on the key points to others
- **C.** Hold the briefings in the local language only, since that is the official language
- **D.** Use translated materials, visual signs and interpreters, and check understanding

**Question 72.** A six-year development has been estimated in today's prices. What should the budget include?

- **A.** Nothing extra, since today's prices are the most accurate
- **B.** A reduction, since prices usually fall over the life of a long project
- **C.** An allowance for expected price increases over the period
- **D.** A large management reserve to replace all estimates

**Question 73.** Midway through the project, the main investor's priority shifts from speed to cost. What should the project manager do?

- **A.** Cut costs immediately in every area of the project without discussion
- **B.** Revisit the plan and communications with the investor to realign them
- **C.** Continue with the original priorities, since they were agreed at the start
- **D.** Ask the other stakeholders whether the investor's priority should be accepted

**Question 74.** At closure, what should the city record about each vendor for future procurements?

- **A.** The vendor's internal salary levels
- **B.** The personal opinions of the project team
- **C.** An evaluation of the vendor's performance
- **D.** The vendor's plans for future products

**Question 75.** Which measure best shows whether a new office building is delivering the value expected by its investor?

- **A.** Occupancy and rental income compared with the business case
- **B.** The number of floors completed on time by the contractor
- **C.** The number of design awards the building has received
- **D.** The final construction cost compared with the budget

**Question 76.** The game engine's vendor will release a major update in the middle of development. What should the team do?

- **A.** Adopt the update on the day it is released, so that the game stays current
- **B.** Assess the update early in a separate branch and plan when to adopt it
- **C.** Ignore the update, since the current version of the engine works well
- **D.** Wait until after launch to look at the update and its possible effects

**Question 77.** In a stable process with normally distributed results, what share of the measurements falls within ±3 standard deviations of the mean?

- **A.** About 50.0%
- **B.** About 68.3%
- **C.** About 99.7%
- **D.** About 95.4%

**Question 78.** The team found 90 defects before a release, and players found 10 more afterwards. What is the defect removal efficiency?

- **A.** 10%
- **B.** 89%
- **C.** 90%
- **D.** 100%

**Question 79.** In a regulated laboratory, a recurring deviation is investigated, and actions are taken both to fix it and to prevent it from happening again. What is this system called?

- **A.** RACI (responsibility assignment)
- **B.** EVM (earned value management)

- **C.** CAPA (corrective and preventive action)
- **D.** WBS (work breakdown structure)

**Question 80.** A junior designer tells the scrum master that the lead designer always dismisses her ideas in team discussions. What should the scrum master do?

- **A.** Report the lead designer's behaviour to the studio's human resources department immediately
- **B.** Ask the lead designer to stop attending the team's design discussions for the next few sprints
- **C.** Tell the junior designer to speak more loudly and confidently in future design discussions
- **D.** Coach the lead designer and use inclusive techniques, such as silent brainstorming

**Question 81.** Who usually reviews unblinded safety data during a clinical trial and can recommend stopping it?

- **A.** The sponsor's marketing department
- **B.** An independent data monitoring committee
- **C.** The trial's project manager alone
- **D.** The patients enrolled in the trial

**Question 82.** A process operating at a six sigma level produces about how many defects per million opportunities?

- **A.** About 3.4
- **B.** About 34
- **C.** About 6
- **D.** About 340

**Question 83.** In a game studio, the creative director wants the game to be as much fun as possible, while the monetisation team pushes for more in-game purchases. Design discussions keep stalling. What should the producer do?

- **A.** Let the creative director decide every design question, since the game's quality depends on it
- **B.** Let the monetisation team decide, since the studio needs revenue to stay in business at all
- **C.** Facilitate agreement on a few design pillars and success measures that balance both views
- **D.** Ask each group to design its own version of the game and test both with players at the end

**Question 84.** Employees resist the new office's shared desks, which were introduced for hybrid working. What should the project do?

- **A.** Announce that desk use will be monitored and reported to the employees' managers
- **B.** Return to fixed desks for everyone, since employees will never accept sharing desks
- **C.** Remove the employees' old desks over a weekend, so that they have no choice but to adapt
- **D.** Explain the reasons, involve employees in designing team zones, and pilot the layout

**Question 85.** In a RACI chart, how many people should be accountable for each deliverable?

- **A.** Everyone involved
- **B.** At least two
- **C.** Nobody, ideally
- **D.** Exactly one

**Question 86.** Late in development, the team learns that the game must ship in 12 languages, but all of its text is hard-coded. What should the product owner do?

- **A.** Postpone localisation until after launch, whatever the publisher's contractual requirements say
- **B.** Ask the translators to work directly in the source code, so that no developer time is needed for it
- **C.** Ship the game in one language only, since the other languages were never formally requested by anyone
- **D.** Add enabler work to externalise the text, prioritised, and treat localisation as a requirement

**Question 87.** Labour law requires consultation with employee representatives before work practices are changed. What should the project do?

- **A.** Implement the change first and consult the representatives afterwards
- **B.** Plan and hold the consultation before implementing the change
- **C.** Consult only the managers, since they will implement the change
- **D.** Avoid the consultation by describing the change as a pilot

**Question 88.** The need for testers rises sharply in the two months before launch. How should the producer plan for this?

- **A.** Hire permanent testers for the peak and keep them on the team afterwards for safety
- **B.** Reduce testing during the peak to match the available testers
- **C.** Ask the developers to do all of the testing in their spare time
- **D.** Plan flexible capacity, such as an external test partner, for the peak

**Question 89.** Two scientists disagree about how to interpret the results of a key experiment, and the disagreement is delaying a decision. What should the project manager do?

- **A.** Repeat the experiment until both scientists reach the same conclusion on their own
- **B.** Report both interpretations to the sponsor and ask the sponsor to choose between the two
- **C.** Accept the interpretation of the more senior scientist, since she has more experience
- **D.** Agree the analysis method together, and involve an independent statistician if needed

**Question 90.** Before tenants can move into a new building, what is normally required?

- **A.** A signed lease from every one of the tenants in the building
- **B.** An occupancy certificate from the authorities
- **C.** The final payment to the architect for the building's design
- **D.** A completed lessons learned session with the whole project team

**Question 91.** Concrete costs are 12% over budget. Analysis shows that the price per cubic metre was as planned, but more concrete was used. What does this indicate?

- **A.** A quantity variance: the team should investigate why more concrete was needed
- **B.** A timing variance that will correct itself automatically before completion of the work
- **C.** A price variance: the supplier has overcharged the project for the concrete
- **D.** An error in the cost reports, since the price was exactly as planned

**Question 92.** The developer wants a single firm to be responsible for both the design and the construction of the building. Which delivery method fits?

- **A.** Design-bid-build

- **B.** Separate design and construction contracts
- **C.** Design-build
- **D.** Construction management at risk

**Question 93.** Which two practices help prevent burnout in a game studio approaching launch? (Choose two.)

- **A.** Making overtime visible and limiting it
- **B.** Planning work based on sustainable capacity
- **C.** Cancelling rest breaks until launch day
- **D.** Hiding overtime from senior management
- **E.** Rewarding the people who work longest

**Question 94.** The team's cycle time is long because its work items are large. What would reduce the cycle time most?

- **A.** Starting more items at the same time
- **B.** Splitting the items into smaller pieces
- **C.** Adding more detail to each item's description
- **D.** Moving to longer sprints

**Question 95.** Who plans the day's work within a sprint?

- **A.** The developers, at the daily scrum
- **B.** The product owner, before each day
- **C.** The sponsor, at the weekly review
- **D.** The scrum master, after each meeting

**Question 96.** Which two actions help identify stakeholders who are not obvious? (Choose two.)

- **A.** Waiting until people complain about the project
- **B.** Reviewing the registers of similar past projects
- **C.** Limiting the list to the organisation's own staff
- **D.** Asking known stakeholders who else is affected
- **E.** Listing only the people named in the charter

**Question 97.** A trial site enrolled a patient before the informed consent form had been signed. What should happen?

- **A.** Report it as a GCP deviation, assess the impact and retrain the site staff
- **B.** Remove the patient from the trial without recording the event, so that the data stays clean
- **C.** Ignore it, since the patient signed the form a few days later without raising any objection
- **D.** Ask the patient to sign the consent form with an earlier date, so that the records are complete

**Question 98.** A biotech company must move its production process from its research lab to a contract manufacturer. How should the knowledge be transferred?

- **A.** By asking the manufacturer to develop its own process from the published research papers
- **B.** A structured technology transfer with documents, training, joint runs and checks
- **C.** By sending one scientist to the manufacturer for a single day to answer its questions
- **D.** By sending the manufacturer the lab notebooks and leaving it to work out the process itself

**Question 99.** Team members complain that one colleague rarely contributes to shared work. What should the scrum master do first?

- **A.** Reassign the colleague's tasks to others, so that the team's progress is protected
- **B.** Report the colleague to the functional manager so that formal action can be taken
- **C.** Ignore the complaints, since self-organising teams must solve such problems alone without help
- **D.** Help the team raise it directly with the colleague, referring to their working agreements

**Question 100.** The project manager drafted the charter for a residential development alone, and the key stakeholders feel no ownership of it. What should the project manager do?

- **A.** Leave the charter as it is, since it has already been written
- **B.** Ask the sponsor to sign the charter without further discussion
- **C.** Revisit the charter with the key stakeholders in a workshop
- **D.** Send the charter to all stakeholders and ask for written approval

**Question 101.** Velocity is stable, but the cycle time of the team's stories is rising. What does this suggest?

- **A.** The team is working faster and finishing its stories sooner than it did in the past
- **B.** Story points have been estimated incorrectly for all of the recent stories
- **C.** Nothing, since velocity is the only measure that matters for agile teams
- **D.** More work is being started in parallel, or stories are waiting longer between steps

**Question 102.** What knowledge must the developer hand over to the building's facility managers at completion?

- **A.** The original design competition entries and all of the rejected design options
- **B.** The marketing brochures that were used to sell the apartments
- **C.** The project team's timesheets and meeting minutes
- **D.** As-built drawings, operation and maintenance manuals, and warranties

**Question 103.** BAC = $2,400,000, and the work is 35% complete. What is the earned value? (Answer in dollars.)

*(Fill in the blank.)*

Answer: _____

**Question 104.** Since the office building was planned, demand in the area has shifted from offices to housing because of remote working. What should the project manager do?

- **A.** Ask the sales team to find office tenants, whatever discounts are needed to fill the building
- **B.** Reassess the business case and the building's mix of uses with the sponsor
- **C.** Continue building offices, since the design has already been approved
- **D.** Stop the project immediately and sell the land as soon as possible

**Question 105.** A clinical trial is stopped early for safety reasons. What should its closure include?

- **A.** Notify the authorities, close the sites, archive the data and record lessons
- **B.** Deleting the trial data, since the trial did not reach its planned end point
- **C.** Continuing to enrol patients until the sponsor makes a final decision
- **D.** Moving the remaining budget to the next trial without any documentation

**Question 106.** Before handover, inspectors walk through each apartment and record the defects that must be fixed. What is this list called?

- **A.** A risk register
- **B.** A requirements matrix
- **C.** A resource calendar
- **D.** A snagging (punch) list

**Question 107.** Other teams frequently break the shared test environment, blocking this team's testing. What should the scrum master do?

- **A.** Raise it as a cross-team impediment and agree environment rules or separate environments
- **B.** Report the other teams to their managers every time the environment is broken
- **C.** Skip testing on the days when the environment has been broken by one of the other teams
- **D.** Ask the team to test only at night, when the other teams are not using the environment

**Question 108.** How should the project communicate with the medicines regulator?

- **A.** Formally, through documented submissions and meetings, via regulatory affairs
- **B.** Informally, through regular phone calls between the scientists and the regulator's staff
- **C.** Through press releases, so that the regulator sees the latest results
- **D.** Through the project's social media accounts, to show transparency

**Question 109.** A project manager coaches a team member by exploring her goal, the current reality, the options available and the way forward. Which coaching model is this?

- **A.** DMAIC
- **B.** ADKAR
- **C.** SMART
- **D.** GROW

**Question 110.** Completion of an apartment building will be three months late, and many buyers' mortgage offers will expire before then. What should the project manager do?

- **A.** Inform the buyers' lenders directly without telling the buyers about the new date
- **B.** Tell the buyers that mortgage offers are their own responsibility, not the developer's
- **C.** Inform the buyers now, explain why and help with options, such as letters for lenders
- **D.** Wait until the delay is certain before informing the buyers, to avoid unnecessary worry

**Question 111.** The building must be ready for the city's inspection by a fixed date. Which type of schedule constraint is this?

- **A.** Start no earlier than
- **B.** Finish no later than
- **C.** Finish-to-start
- **D.** As soon as possible

**Question 112.** Before closing the construction contract, what must be agreed with the main contractor?

- **A.** The design of the building's next phase
- **B.** The marketing plan for selling the remaining apartments
- **C.** The contractor's staffing plan for its next project

- **D.** The final account, including all variations and claims

**Question 113.** Marketing wants to show unfinished features in a launch trailer as if they were final, although some may be cut. What should the product owner do?

- **A.** Ask the team to finish all of the features shown, whatever happens to the release date
- **B.** Allow the trailer as planned, since trailers always show an idealised version of the game
- **C.** Leave the decision to marketing, since trailers are not part of the product's scope
- **D.** Agree honest messaging: show only features likely to ship, or label them as in progress

**Question 114.** Privacy advocates publicly criticise the smart city sensors, claiming that they will track citizens. What should the project manager do?

- **A.** Remove all sensors from the plan to avoid any further criticism
- **B.** Engage them early, share the privacy design and invite their review
- **C.** Ignore them, since the sensors comply with the data protection law
- **D.** Ask the city's lawyers to send them a formal warning about their claims

**Question 115.** The issue log has grown to 80 open issues, and many have not been updated for weeks. What should the project manager do?

- **A.** Review and prioritise the issues regularly with their owners, closing or escalating them
- **B.** Delete all of the issues that are more than a month old to make the log manageable
- **C.** Ask the sponsor to deal with the issues, since there are too many for the team
- **D.** Add a new column to the log so that the open issues can be sorted and filtered more easily

**Question 116.** Workers discover asbestos during the refurbishment of an old warehouse. What should the project manager do?

- **A.** Stop work in the area and follow the legal requirements for removal by licensed specialists
- **B.** Continue the work carefully, since the asbestos is only a small amount in one area of the building
- **C.** Cover the asbestos with new panels so that the refurbishment can continue
- **D.** Ask the workers to remove the asbestos themselves to save time and money

**Question 117.** Buyers of apartments sold before construction expect premium finishes, but the sales contracts specify standard finishes. What should the project manager do?

- **A.** Wait until handover and let the buyers raise any concerns at that point
- **B.** Install premium finishes in all apartments to avoid complaints from buyers
- **C.** Tell the buyers that nothing can be discussed once the contracts are signed
- **D.** Clarify the specification with the buyers and offer upgrades as paid changes

**Question 118.** The amount of rock to be excavated is uncertain, so the buyer will pay a fixed rate for each cubic metre removed. Which contract type is this?

- **A.** Unit price
- **B.** Firm fixed price
- **C.** Time and materials
- **D.** Cost plus fixed fee

**Question 119.** Which practice automatically checks code for common defects and style problems without running it?

- **A.** Static code analysis
- **B.** Exploratory testing
- **C.** User acceptance testing
- **D.** Load testing

**Question 120.** The trial team wants to change the dosing schedule set out in the study protocol. What must happen before the change is implemented?

- **A.** The change is tried at one site first to see whether it works in practice
- **B.** A protocol amendment is approved by the regulator and the ethics committee
- **C.** Nothing, since the trial team owns the protocol and can change it at any time
- **D.** The change is announced to the trial sites by email from the project manager

> ☕ **Optional 10-minute break.** In the real exam, once you start a break you cannot return to the questions of the previous section, so review flagged questions before you go.

## Section 3 — Questions 121 to 180

**Question 121.** Activity A (5 days) starts the project. B (7 days) and C (4 days) both follow A. D (6 days) follows C. E (3 days) follows both B and D. What is the duration of the project, in days? (Answer with a number.)

*(Fill in the blank.)*

Answer: _____

**Question 122.** Developers in three studios in different countries identify with their own studio rather than with the game they are building together. What should the producer do?

- **A.** Create shared goals and rituals around the game, such as joint demos and a common team charter
- **B.** Let each studio keep its own goals, since local identity motivates the developers in each studio
- **C.** Rank the studios against each other each month, so that competition drives them to perform better
- **D.** Merge the three studios into a single location, so that everyone works in the same building

**Question 123.** Staff from a contract research organisation will join the clinical trial team. What must happen before they start work?

- **A.** They are trained on the study's procedures and their qualifications are confirmed
- **B.** Nothing, since the contract research organisation is responsible for its own staff
- **C.** They read the trial protocol in their own time and start work the next day
- **D.** They start work immediately and learn the procedures while they work on the trial

**Question 124.** A drug candidate has a 20% chance of success. If it succeeds, its NPV will be $500 million. The development cost is $60 million. What is its risk-adjusted value? (Answer in millions of dollars, as a number.)

*(Fill in the blank.)*

Answer: _____

**Question 125.** The team must cut 20% of the scope to keep the launch date. Which items should be removed first?

- **A.** Those at the bottom of the original feature list
- **B.** Those with the lowest value relative to their effort
- **C.** Those requested by the most junior stakeholders
- **D.** Those that the developers find least interesting to build

**Question 126.** Which practice spreads specialised knowledge, such as shader programming, across the studio's teams at low cost?

- **A.** Keeping the specialists in a separate expert team
- **B.** Hiring an external trainer for each team every month
- **C.** Regular internal tech talks, recorded for later viewing
- **D.** Asking specialists to write a manual once a year

**Question 127.** To finish on time, the site must complete one apartment floor every five days. What is this planned pace of production called?

- **A.** Takt time
- **B.** Lag time
- **C.** Lead time
- **D.** Float time

**Question 128.** The power company's connection to the new building is late, so the building cannot be commissioned. What should the project manager do?

- **A.** Start the commissioning without power, testing only the systems that do not need it
- **B.** Ask the contractor to connect the building to the grid without the power company
- **C.** Wait for the connection, since the power company is outside the project's control
- **D.** Escalate with the power company and plan workarounds, such as generators

**Question 129.** Each approved change request was small, but together they have used up most of the contingency reserve. What should the project manager do?

- **A.** Stop accepting any further change requests from stakeholders until the project ends
- **B.** Report the cumulative impact and review the remaining reserve and priorities with the sponsor
- **C.** Continue approving small changes, since each one is within the project manager's authority
- **D.** Use the management reserve to replace the contingency reserve, without informing anyone about it

**Question 130.** Which two measures best show the value of new smart traffic lights? (Choose two.)

- **A.** The share of the budget spent so far
- **B.** Vehicle emissions compared with a baseline
- **C.** Average journey times compared with a baseline
- **D.** The number of project meetings held
- **E.** The number of traffic lights installed

**Question 131.** A change request proposes adding licence plate recognition to the parking sensors. What should happen before the change control board decides?

- **A.** A full impact assessment, including the impact on data protection
- **B.** An approval by the vendor, since it will build the new feature
- **C.** Nothing, since the feature would make parking enforcement easier
- **D.** A trial on the streets to see how citizens react to the new feature before deciding

**Question 132.** Trial sites enter their data late, which delays the analysis. What should the project manager do?

- **A.** Replace the sites that are late with new sites, whatever that costs the project
- **B.** Ask the data managers to estimate the missing values until the sites catch up
- **C.** Wait until the end of the trial and then ask the sites to enter all of the data
- **D.** Work with the sites to remove the barriers, such as complex forms

**Question 133.** How can a game studio engage its player community during development?

- **A.** By reading the reviews after launch and fixing the issues that players mention most

- **B.** Through community channels and early-access builds, with feedback feeding the backlog
- **C.** By keeping the game secret until launch, so that competitors cannot copy its features
- **D.** By asking only the studio's own staff to play the game and report their impressions

**Question 134.** A vendor's delay means the smart city app will not be ready for a public event at which the city's communications team has already promised to launch it. What should the project manager do?

- **A.** Inform the communications team now and agree revised messaging and options, such as a preview
- **B.** Ask the mayor to postpone the public event without explaining the reason to the city's staff
- **C.** Ask the vendor to release the app on the promised date, even if it has not been fully tested
- **D.** Let the communications team find out at the final rehearsal, when the situation is certain

**Question 135.** Which two of the following are baselines? (Choose two.)

- **A.** The issue log
- **B.** The risk register
- **C.** The schedule baseline
- **D.** The stakeholder register
- **E.** The cost baseline

**Question 136.** The developer and a buyer disagree about what "substantial completion" of the apartment means. What should the project manager do?

- **A.** Use the developer's interpretation, since the developer wrote the contract
- **B.** Let the sales agent decide, since the agent sold the apartment to the buyer
- **C.** Use the contract's definition and the agreed evidence, such as an occupancy permit
- **D.** Accept the buyer's interpretation, since the buyer is the paying customer and must be satisfied

**Question 137.** A competitor's drug, which works in a similar way, has just failed in clinical trials, and the biotech team starts to doubt its own project. What should the project manager do?

- **A.** Ask the sponsor to announce bonuses, so that the team stays motivated despite the news
- **B.** Tell the team that the competitor's failure is irrelevant and that no discussion is needed
- **C.** Stop the project immediately, since a similar drug has already failed in the clinic
- **D.** Review with the team how this project differs, using the data, and reaffirm the vision

**Question 138.** Server costs for the live game rise sharply with the number of players. How should the team control these costs?

- **A.** Move all servers to the cheapest provider without assessing performance or reliability
- **B.** Monitor the cost per player and optimise the architecture, with alerts on spending
- **C.** Pay the bills as they arrive, since more players also bring in more revenue
- **D.** Limit the number of players who can join, so that server costs stay within the budget

**Question 139.** Which technique best checks whether players find a new level fun and easy to understand?

- **A.** A load test of the servers that host the game's levels
- **B.** A code review of the level's scripts by senior developers
- **C.** Playtesting with real players, observing them as they play

- **D.** A survey of the designers who created the new level

**Question 140.** The regulator introduces an accelerated approval pathway for drugs like the team's. What should the project manager do?

- **A.** Continue with the standard pathway, since it was planned at the start
- **B.** Wait until a competitor has used the new pathway successfully before considering it
- **C.** Apply for the accelerated pathway at once, without any assessment
- **D.** Assess whether the drug is eligible and how the pathway would change the plan

**Question 141.** Before a biotech company buys a new laboratory instrument, which document should define what the instrument must do?

- **A.** The laboratory's lessons learned register
- **B.** A user requirements specification
- **C.** The supplier's product brochure
- **D.** The instrument's maintenance log

**Question 142.** A national innovation fund co-finances 40% of a project on condition that the project submits quarterly reports and publishes its data openly. How should these conditions be handled?

- **A.** Address them at the end of the project, when all of the data is available
- **B.** Ask the fund to remove the conditions, since they create extra work
- **C.** Build them into the governance framework and the plan, with clear owners
- **D.** Treat them as optional, since the city provides most of the funding

**Question 143.** A trial needs 120 patients. After six months, 36 have been enrolled at a steady rate of 6 per month. At this rate, how many more months will enrolment take?

- **A.** 8 months
- **B.** 10 months
- **C.** 14 months
- **D.** 20 months

**Question 144.** Several parallel paths converge into a single milestone, making the milestone more likely to slip than any single path suggests. What is this effect called?

- **A.** Merge bias
- **B.** Gold plating
- **C.** Scope creep
- **D.** Resource smoothing

**Question 145.** A publisher funds a game in milestones and must approve each one. How can the studio stay agile within this governance?

- **A.** Ignore the milestones, since agile teams should not be constrained by external governance
- **B.** Stop using sprints and follow a detailed predictive plan agreed with the publisher at the start
- **C.** Ask the publisher to approve every sprint plan and every change to the backlog in advance
- **D.** Agree outcome-based milestone criteria, such as playable builds, and plan freely in between

**Question 146.** The agile coach wants teams to assess areas such as speed, learning and fun regularly and visually, so that trends become visible. What should the coach introduce?

- **A.** A monthly velocity ranking
- **B.** A quarterly audit by the PMO
- **C.** An annual employee survey
- **D.** A regular team health check

**Question 147.** A buyer asks for changes to the layout of his apartment after the design freeze. What should the project manager do?

- **A.** Refuse without explanation, since the design freeze applies to every buyer in the building
- **B.** Make the changes informally on site, so that the formal change process can be avoided
- **C.** Explain the freeze and offer options, with cost and timing, through the change process
- **D.** Make the changes free of charge, since the buyer is an important customer for the developer

**Question 148.** The team wants to release a risky new feature but be able to switch it off immediately if problems appear. What should it use?

- **A.** A longer sprint
- **B.** A feature toggle
- **C.** A larger team
- **D.** A change freeze

**Question 149.** Analysis shows that the project will miss a milestone unless more crews are added, which requires extra budget. What should happen next?

- **A.** Ask the crews to work unpaid overtime
- **B.** Raise a change request with the analysis and options
- **C.** Add the crews immediately and report the extra cost at the end of the project
- **D.** Accept the delay without informing anyone

**Question 150.** Each enrolled patient costs $25,000, and the budget assumed 120 patients. A protocol amendment increases the number of patients to 150. What is the additional cost?

- **A.** $600,000
- **B.** $750,000
- **C.** $900,000
- **D.** $3,750,000

**Question 151.** Which tool helps the project manager store plans, schedules and documents and track performance in one place?

- **A.** A communications management plan
- **B.** A project management information system
- **C.** A lessons learned register
- **D.** A stakeholder engagement matrix

**Question 152.** Another city has already deployed similar sensors. How can the project benefit from its experience?

- **A.** Through site visits and peer exchange before finalising the design

- **B.** By waiting to see whether the other city's project eventually fails
- **C.** By copying the other city's design without any further discussion
- **D.** By hiring the other city's vendor without asking about its results

**Question 153.** City councillors approve the smart city budget but have very little time. How should the project engage them?

- **A.** Through the vendors, who can explain the technology better
- **B.** Only at the end of the project, when the results are available
- **C.** With short, outcome-focused briefings at key decision points
- **D.** With the full technical documentation sent before each meeting

**Question 154.** Stakeholders add 60 points of new scope to the release backlog. What should the product owner do?

- **A.** Reject the new scope, since the release backlog was agreed at the start of the release
- **B.** Accept the new scope and ask the team to increase its velocity to absorb the extra work
- **C.** Add the new scope to the next sprint, whatever was already planned for that sprint
- **D.** Show the impact on the release forecast and agree trade-offs with stakeholders

**Question 155.** Artists resist moving to a new game engine because they are experts in the old tools. What should the studio do?

- **A.** Provide training and time to learn, with senior artists as mentors
- **B.** Switch to the new engine immediately and require the artists to adapt within a week
- **C.** Let each artist decide whether to use the old or the new engine on the project
- **D.** Replace the artists who resist with new artists who already know the new engine

**Question 156.** Five city departments have overlapping requirements for the new platform. Which technique brings them together most efficiently?

- **A.** A facilitated requirements workshop
- **B.** A review of each department's annual report
- **C.** An interview with each department's head only
- **D.** A separate questionnaire for each department

**Question 157.** Which recognition practice fits a self-organising team best?

- **A.** Recognition only at the end of the project, in the final report
- **B.** Bonuses for the individuals who complete the most story points
- **C.** An employee-of-the-month award chosen by the project manager alone after each sprint
- **D.** Peer-to-peer recognition, for example thanking colleagues in retrospectives

**Question 158.** Risk X has a 25% probability of causing an 8-day delay. Risk Y has a 50% probability of causing a 4-day delay. What is their combined expected schedule impact, in days? (Answer with a number.)

*(Fill in the blank.)*

Answer: _____

**Question 159.** Investors expect a new drug to reach the market within three years, but regulatory approval typically takes much longer. What should the project manager do?

- **A.** Avoid the subject until the first clinical trial results become available
- **B.** Promise the three-year timeline to keep the investors' support for the work
- **C.** Plan to skip some regulatory steps so that the timeline can still be met
- **D.** Explain the regulatory steps and timelines, and agree realistic milestones

**Question 160.** The team sizes epics as small, medium or large and knows the average cost of each size from past work. What estimating approach is this?

- **A.** Bottom-up estimating of every task
- **B.** Three-point estimating using PERT for each epic
- **C.** Analogous estimating based on relative size
- **D.** Parametric estimating using a cost formula

**Question 161.** Retail tenants assume that the developer will install their shop fittings, but the lease covers only "shell and core". Where should the boundary between the developer's work and the tenants' work be defined?

- **A.** In the minutes of the first meeting with each of the tenants
- **B.** Nowhere, since it can be agreed on site as the work progresses
- **C.** In the scope statement and the tenants' lease specification
- **D.** In the developer's marketing brochure for the retail space

**Question 162.** Some department heads publicly support the new process but tell their teams privately to keep using the old one. What should the project manager do?

- **A.** Report the department heads to the board without speaking to them first
- **B.** Tell the teams to ignore their department heads' private instructions and follow the process
- **C.** Raise it with the sponsor and the leaders to align their messages and behaviour
- **D.** Ignore it, since the department heads support the process in public

**Question 163.** The organisation's projects consistently underestimate their durations. Which technique adjusts estimates using the actual outcomes of similar past projects?

- **A.** Reference class forecasting
- **B.** Resource smoothing
- **C.** Rolling wave planning
- **D.** Critical chain buffering

**Question 164.** The game uses licensed middleware. Which closure task is easy to overlook when development ends and live operations begin?

- **A.** Deleting the middleware from the build, since development has now finished
- **B.** Asking the middleware vendor to support the game free of charge from now on
- **C.** Cancel the middleware licences, since the game has now been released to players
- **D.** Confirming the licence terms and transferring the licences for live operations

**Question 165.** Only two console development kits are available for three teams, and the teams often wait for access. What should the scrum master do?

- **A.** Ask one of the teams to stop all console work until more kits arrive
- **B.** Let the team that arrives first each morning use the kits for the whole of that day
- **C.** Accept the waiting, since development kits are expensive and hard to get
- **D.** Escalate the need for more kits, and agree a fair booking system in the meantime

**Question 166.** The WBS for a new building includes all of the work except landscaping, which the team considered "obvious". Which principle has been violated?

- **A.** The 100% rule
- **B.** Rolling wave planning
- **C.** The 8/80 rule
- **D.** Progressive elaboration

**Question 167.** Before sharing confidential data with potential contract research organisations during selection, what should be in place?

- **A.** Non-disclosure agreements
- **B.** A letter of intent to each one
- **C.** A signed final contract
- **D.** A press release about the trial

**Question 168.** Match each threat response with its description.

*(Matching: pair every numbered item with one lettered option.)*

| Item | Option |
|---|---|
| **1.** Escalate | **A.** Change the plan to eliminate the threat |
| **2.** Avoid | **B.** Pass the threat to a higher level that owns it |
| **3.** Transfer | **C.** Shift the impact and ownership to a third party |
| **4.** Mitigate | **D.** Reduce the probability or the impact of the threat |

**Question 169.** The launch date is fixed. How should the release plan be structured?

- **A.** All of the planned scope, based on the team's best sprint so far this year
- **B.** A list of features with an exact delivery date for each of them
- **C.** Must-have scope within a cautious forecast, with stretch items marked
- **D.** No release plan, since agile teams do not plan beyond one sprint

**Question 170.** What does a project's internal rate of return (IRR) represent?

- **A.** The discount rate at which the project's net present value equals zero
- **B.** The interest rate charged by the bank that finances the project
- **C.** The time it takes for the project to pay back its investment
- **D.** The total profit of the project divided by its total cost

**Question 171.** Which two conflict resolution approaches usually provide only temporary relief rather than resolving the underlying problem? (Choose two.)

- **A.** Confronting the problem
- **B.** Consensus building
- **C.** Withdrawing (avoiding)
- **D.** Smoothing (accommodating)
- **E.** Collaborating (problem solving)

**Question 172.** The city needs an innovative solution that it cannot fully specify in advance, and public procurement rules apply. What should it do?

- **A.** Use a procedure that allows dialogue with bidders, within the rules
- **B.** Write a detailed specification anyway and award the contract on the lowest price
- **C.** Award the contract directly to a vendor the city trusts, without any competition
- **D.** Delay the procurement until the city can specify the solution in full detail

**Question 173.** Traffic managers expect live data from the new platform, but it updates only every 15 minutes. What should the project manager do?

- **A.** Clarify the update interval early and agree the requirement based on their real needs
- **B.** Say nothing, since 15 minutes is close enough to live data for most traffic decisions
- **C.** Promise live data in the next release, without checking whether it is technically possible
- **D.** Ask the traffic managers to adapt their work to the platform's 15-minute update cycle

**Question 174.** Six months into a transformation, enthusiasm is fading. Which two actions help sustain momentum? (Choose two.)

- **A.** Making short-term wins visible and celebrating them
- **B.** Disbanding the group leading the change
- **C.** Reducing communication about the change
- **D.** Using early successes to drive further improvements
- **E.** Declaring the transformation complete

**Question 175.** During the final three months before the hospital opens, the project manager brings key team members together in one room. Why?

- **A.** To speed up communication and decisions during a critical period
- **B.** To monitor how many hours each team member works every day
- **C.** To prevent team members from talking to other stakeholders
- **D.** To reduce the cost of office space during the final months

**Question 176.** New subcontractors join a large building project every few weeks, and many do not know the project's objectives or site rules. What should the project manager introduce?

- **A.** A rule that subcontractors may start work only after the main contractor's training course
- **B.** A monthly meeting where the project manager presents the latest status to all subcontractors
- **C.** A copy of the full project management plan for each subcontractor to read on arrival
- **D.** A structured induction covering the project's objectives, values and safety rules

**Question 177.** Some countries regulate paid random rewards ("loot boxes") in games. What should the product owner do?

- **A.** Wait for regulators to object to the features before making any changes to them
- **B.** Remove all in-game purchases from the game to avoid any regulation in any market
- **C.** Check the rules in each target market with legal experts and adapt the features per market
- **D.** Use the same features in every market, since the game is developed and published from one country

**Question 178.** Instead of each developer working on a separate story, the whole team works together on the top-priority story until it is done. What is this called?

- **A.** Crashing
- **B.** Multitasking
- **C.** Gold plating
- **D.** Swarming

**Question 179.** Before development starts, the product owner asks the team to write the press release that will announce the finished product to customers. What is the purpose of this exercise?

- **A.** To align the team on the customer outcome, working backwards from it
- **B.** To check whether the team's writing skills are good enough
- **C.** To fix the scope at the start so that it cannot change during development
- **D.** To prepare the marketing materials early and save time at launch

**Question 180.** Progress meetings with the contractors are long and rarely produce clear actions. What should the project manager do?

- **A.** Hold the meetings more often, so that each one has fewer topics to discuss and ends sooner
- **B.** Send an agenda in advance, timebox the items and record actions with owners and dates
- **C.** Cancel the meetings and ask the contractors to send written reports instead
- **D.** Invite more people to the meetings, so that decisions can be made on the spot

## Answer Key

| Q | Answer | Domain | Task | Approach | Q | Answer | Domain | Task | Approach |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **B** | Process | PR1 | Predictive | 91 | **A** | Process | PR6 | Hybrid |
| 2 | **D** | People | PE6 | Hybrid | 92 | **C** | Process | PR5 | Predictive |
| 3 | **A** | People | PE5 | Agile | 93 | **A, B** | People | PE3 | Agile |
| 4 | **C** | People | PE4 | Agile | 94 | **B** | Process | PR8 | Agile |
| 5 | **C** | Bus. Env. | BE2 | Predictive | 95 | **A** | Process | PR1 | Agile |
| 6 | **B** | Bus. Env. | BE8 | Predictive | 96 | **B, D** | People | PE4 | Hybrid |
| 7 | **D** | Process | PR2 | Agile | 97 | **A** | Bus. Env. | BE2 | Hybrid |
| 8 | **1→D, 2→A, 3→C, 4→B** | Process | PR8 | Predictive | 98 | **B** | People | PE7 | Predictive |
| 9 | **C** | Bus. Env. | BE3 | Predictive | 99 | **D** | People | PE3 | Agile |
| 10 | **D** | People | PE8 | Hybrid | 100 | **C** | People | PE1 | Predictive |
| 11 | **B, C** | Process | PR7 | Hybrid | 101 | **D** | Process | PR9 | Agile |
| 12 | **D** | Bus. Env. | BE4 | Hybrid | 102 | **D** | People | PE7 | Predictive |
| 13 | **C** | Bus. Env. | BE2 | Hybrid | 103 | **$840,000** | Process | PR6 | Predictive |
| 14 | **B** | Bus. Env. | BE5 | Hybrid | 104 | **B** | Bus. Env. | BE8 | Hybrid |
| 15 | **C** | Bus. Env. | BE7 | Hybrid | 105 | **A** | Process | PR10 | Hybrid |
| 16 | **B** | Bus. Env. | BE8 | Predictive | 106 | **D** | Process | PR7 | Predictive |
| 17 | **A** | Process | PR5 | Hybrid | 107 | **A** | Bus. Env. | BE4 | Agile |
| 18 | **B** | People | PE4 | Predictive | 108 | **A** | People | PE8 | Predictive |
| 19 | **B** | Process | PR2 | Hybrid | 109 | **D** | People | PE3 | Hybrid |
| 20 | **B** | People | PE2 | Predictive | 110 | **C** | People | PE6 | Predictive |
| 21 | **A, D** | People | PE1 | Agile | 111 | **B** | Process | PR8 | Predictive |
| 22 | **C** | Bus. Env. | BE1 | Predictive | 112 | **D** | Process | PR10 | Predictive |

| Q | Answer | Domain | Task | Approach | Q | Answer | Domain | Task | Approach |
|---|---|---|---|---|---|---|---|---|---|
| 23 | **A** | Process | PR9 | Hybrid | 113 | **D** | People | PE5 | Agile |
| 24 | **A** | Process | PR1 | Hybrid | 114 | **B** | People | PE4 | Hybrid |
| 25 | **B** | Bus. Env. | BE6 | Agile | 115 | **A** | Bus. Env. | BE4 | Predictive |
| 26 | **B** | Bus. Env. | BE1 | Agile | 116 | **A** | Bus. Env. | BE2 | Predictive |
| 27 | **C** | Process | PR1 | Agile | 117 | **D** | People | PE5 | Predictive |
| 28 | **B** | Process | PR9 | Predictive | 118 | **A** | Process | PR5 | Predictive |
| 29 | **A** | People | PE5 | Hybrid | 119 | **A** | Process | PR7 | Agile |
| 30 | **C** | Process | PR6 | Predictive | 120 | **B** | Bus. Env. | BE3 | Hybrid |
| 31 | **B** | People | PE2 | Predictive | 121 | **18** | Process | PR8 | Predictive |
| 32 | **B** | Process | PR1 | Hybrid | 122 | **A** | People | PE1 | Hybrid |
| 33 | **B** | People | PE4 | Predictive | 123 | **A** | Process | PR4 | Hybrid |
| 34 | **B** | People | PE3 | Agile | 124 | **40** | Process | PR3 | Predictive |
| 35 | **B** | Process | PR9 | Predictive | 125 | **B** | Process | PR3 | Agile |
| 36 | **C** | Process | PR5 | Agile | 126 | **C** | People | PE7 | Agile |
| 37 | **C** | Process | PR10 | Hybrid | 127 | **A** | Process | PR8 | Hybrid |
| 38 | **D** | Bus. Env. | BE5 | Hybrid | 128 | **D** | Bus. Env. | BE4 | Predictive |
| 39 | **A** | Bus. Env. | BE8 | Agile | 129 | **B** | Bus. Env. | BE3 | Hybrid |
| 40 | **C** | People | PE8 | Agile | 130 | **B, C** | Process | PR3 | Hybrid |
| 41 | **C** | People | PE6 | Agile | 131 | **A** | Bus. Env. | BE3 | Predictive |
| 42 | **B** | People | PE2 | Hybrid | 132 | **D** | Bus. Env. | BE4 | Hybrid |
| 43 | **A** | Process | PR7 | Hybrid | 133 | **B** | People | PE4 | Agile |
| 44 | **D** | People | PE3 | Predictive | 134 | **A** | People | PE6 | Hybrid |
| 45 | **D** | Bus. Env. | BE2 | Agile | 135 | **C, E** | Process | PR1 | Predictive |
| 46 | **B** | Process | PR6 | Predictive | 136 | **C** | People | PE5 | Predictive |
| 47 | **C** | People | PE4 | Predictive | 137 | **D** | People | PE1 | Hybrid |
| 48 | **B** | Bus. Env. | BE1 | Hybrid | 138 | **B** | Process | PR6 | Agile |
| 49 | **C** | Process | PR8 | Agile | 139 | **C** | Process | PR7 | Agile |

| Q | Answer | Domain | Task | Approach | Q | Answer | Domain | Task | Approach |
|---|---|---|---|---|---|---|---|---|---|
| 50 | **C** | People | PE2 | Agile | 140 | **D** | Bus. Env. | BE8 | Hybrid |
| 51 | **B** | People | PE2 | Predictive | 141 | **B** | Process | PR2 | Predictive |
| 52 | **A** | Process | PR4 | Predictive | 142 | **C** | Bus. Env. | BE1 | Predictive |
| 53 | **A, D** | People | PE8 | Hybrid | 143 | **C** | Process | PR9 | Predictive |
| 54 | **B** | People | PE7 | Agile | 144 | **A** | Process | PR8 | Predictive |
| 55 | **D, E** | Bus. Env. | BE5 | Hybrid | 145 | **D** | Bus. Env. | BE1 | Agile |
| 56 | **A** | People | PE6 | Agile | 146 | **D** | Bus. Env. | BE6 | Agile |
| 57 | **A** | People | PE3 | Predictive | 147 | **C** | People | PE6 | Predictive |
| 58 | **C** | People | PE8 | Agile | 148 | **B** | Bus. Env. | BE5 | Agile |
| 59 | **D** | Process | PR8 | Predictive | 149 | **B** | Process | PR9 | Hybrid |
| 60 | **D** | Process | PR7 | Agile | 150 | **B** | Process | PR6 | Hybrid |
| 61 | **D** | Process | PR5 | Hybrid | 151 | **B** | Process | PR1 | Predictive |
| 62 | **C** | Process | PR3 | Agile | 152 | **A** | People | PE7 | Hybrid |
| 63 | **D** | People | PE7 | Hybrid | 153 | **C** | People | PE4 | Hybrid |
| 64 | **C** | Process | PR3 | Hybrid | 154 | **D** | Bus. Env. | BE3 | Agile |
| 65 | **D** | Bus. Env. | BE3 | Predictive | 155 | **A** | Bus. Env. | BE7 | Agile |
| 66 | **D** | Process | PR2 | Agile | 156 | **A** | Process | PR2 | Hybrid |
| 67 | **1→B, 2→D, 3→A, 4→C** | People | PE3 | Predictive | 157 | **D** | People | PE3 | Agile |
| 68 | **C** | Process | PR3 | Agile | 158 | **4** | Bus. Env. | BE5 | Predictive |
| 69 | **A** | Process | PR4 | Agile | 159 | **D** | People | PE5 | Predictive |
| 70 | **C** | People | PE3 | Hybrid | 160 | **C** | Process | PR6 | Agile |
| 71 | **D** | People | PE8 | Predictive | 161 | **C** | Process | PR2 | Predictive |
| 72 | **C** | Process | PR6 | Predictive | 162 | **C** | Bus. Env. | BE7 | Hybrid |
| 73 | **B** | People | PE6 | Predictive | 163 | **A** | Bus. Env. | BE6 | Hybrid |
| 74 | **C** | Process | PR10 | Predictive | 164 | **D** | Process | PR10 | Agile |

| Q | Answer | Domain | Task | Approach | Q | Answer | Domain | Task | Approach |
|---|---|---|---|---|---|---|---|---|---|
| 75 | **A** | Process | PR3 | Predictive | 165 | **D** | Bus. Env. | BE4 | Agile |
| 76 | **B** | Bus. Env. | BE5 | Agile | 166 | **A** | Process | PR2 | Predictive |
| 77 | **C** | Process | PR7 | Predictive | 167 | **A** | Process | PR5 | Predictive |
| 78 | **C** | Process | PR9 | Agile | 168 | **1→B, 2→A, 3→C, 4→D** | Bus. Env. | BE5 | Predictive |
| 79 | **C** | Bus. Env. | BE6 | Hybrid | 169 | **C** | Process | PR1 | Agile |
| 80 | **D** | People | PE2 | Agile | 170 | **A** | Process | PR3 | Predictive |
| 81 | **B** | Bus. Env. | BE1 | Hybrid | 171 | **C, D** | People | PE2 | Predictive |
| 82 | **A** | Bus. Env. | BE6 | Predictive | 172 | **A** | Process | PR5 | Hybrid |
| 83 | **C** | People | PE1 | Agile | 173 | **A** | People | PE5 | Hybrid |
| 84 | **D** | Bus. Env. | BE7 | Predictive | 174 | **A, D** | Bus. Env. | BE7 | Agile |
| 85 | **D** | Process | PR4 | Predictive | 175 | **A** | Process | PR4 | Predictive |
| 86 | **D** | Process | PR2 | Agile | 176 | **D** | People | PE1 | Predictive |
| 87 | **B** | Bus. Env. | BE7 | Predictive | 177 | **C** | Bus. Env. | BE2 | Agile |
| 88 | **D** | Process | PR4 | Hybrid | 178 | **D** | Process | PR4 | Agile |
| 89 | **D** | People | PE2 | Hybrid | 179 | **A** | People | PE1 | Agile |
| 90 | **B** | Process | PR10 | Predictive | 180 | **B** | People | PE8 | Predictive |

## Detailed Explanations

**Q1 — Answer: B** · *Process · PR1 Develop an integrated project management plan and plan delivery · Predictive*

*Correct:* B. In the calendars used for scheduling

Project and resource calendars define when work can be done. Supplier shutdowns are reflected so that the schedule is realistic.

**Q2 — Answer: D** · *People · PE6 Manage stakeholder expectations · Hybrid*

*Correct:* D. Investigate quickly, fix or suspend the faulty function and put things right

When a delivered solution harms stakeholders, the project acts quickly to investigate, stop the harm, correct it and communicate openly.

**Q3 — Answer: A** · *People · PE5 Align stakeholder expectations · Agile*

*Correct:* A. Communicate a realistic update cadence and roadmap, and keep to it

A realistic, reliable cadence builds more trust than ambitious promises that cannot be kept.

**Q4 — Answer: C** · *People · PE4 Engage stakeholders · Agile*

*Correct:* C. Bring player data and tests into decisions, while keeping the CEO engaged

Evidence helps counter the "highest-paid person's opinion" effect, while respectful engagement keeps a powerful stakeholder involved.

**Q5 — Answer: C** · *Business Environment · BE2 Plan and manage project compliance · Predictive*

*Correct:* C. Build it into the design requirements and verify it at reviews and before handover

Regulatory requirements are built into the requirements and verified progressively, which avoids expensive late changes.

**Q6 — Answer: B** · *Business Environment · BE8 Evaluate external business environment changes · Predictive*

*Correct:* B. Monitor their finances, identify alternatives and consider protections such as bonds

Economic conditions create supply chain risks. Monitoring, contingency plans and contractual protections reduce the impact of a subcontractor failing.

**Q7 — Answer: D** · *Process · PR2 Develop and manage project scope · Agile*

*Correct:* D. Apply the content lock and add the new ideas to the post-launch backlog

A content lock protects the launch. New ideas are not lost: they are captured in the backlog for later releases.

**Q8 — Answer: 1→D, 2→A, 3→C, 4→B** · *Process · PR8 Plan and manage schedule · Predictive*

*Correct pairs:* 1. Mandatory → Required by the contract or the nature of the work; 2. Discretionary → Based on best practice or the team's preference; 3. External → Involves activities outside the project; 4. Internal → Involves activities within the project's control

Dependency types help the project manager decide which sequences can be changed, for example when compressing the schedule.

**Q9 — Answer: C** · *Business Environment · BE3 Manage and control changes · Predictive*

*Correct:* C. A variation order agreed under the contract

Contract changes in construction are made through variation (change) orders, which record the agreed scope, cost and time impacts.

**Q10 — Answer: D** · *People · PE8 Plan and manage communication · Hybrid*

*Correct:* D. Single points of contact and agreed communication protocols

Clear points of contact and protocols reduce noise, avoid conflicting instructions and keep information flowing reliably.

**Q11 — Answer: B, C** · *Process · PR7 Plan and optimize quality of products/deliverables · Hybrid*

*Correct:* B. Training staff in the procedures before they start work / C. Designing the process so that it consistently produces quality

Prevention builds quality into the process and the people. Inspections and complaint analysis detect defects after they have occurred.

**Q12 — Answer: D** · *Business Environment · BE4 Remove impediments and manage issues · Hybrid*

*Correct:* D. Escalate it to programme management so that it can be handled once for all projects

Issues that affect several projects are best handled at programme level, where a coordinated response is possible.

**Q13 — Answer: C** · *Business Environment · BE2 Plan and manage project compliance · Hybrid*

*Correct:* C. Complete the DPIA early and design the analytics for anonymous counting only

Privacy requirements are built in from the start: the DPIA identifies risks, and the design limits processing to what the guidance allows.

**Q14 — Answer: B** · *Business Environment · BE5 Plan and manage risk · Hybrid*

*Correct:* B. Build security into the design and testing, and prepare a tested incident response plan

Mitigation through security by design, testing and a rehearsed incident response reduces both the probability and the impact. Contracts cannot transfer the city's responsibility for its services.

**Q15 — Answer: C** · *Business Environment · BE7 Support organizational change · Hybrid*

*Correct:* C. How the change benefits them, such as safer work and less patrolling in bad weather

People adopt change more readily when they see what is in it for them. Threats and technical arguments do little to build desire.

**Q16 — Answer: B** · *Business Environment · BE8 Evaluate external business environment changes · Predictive*

*Correct:* B. Assess the legal impact, engage the vendor, identify alternatives and use change control

New regulations are analysed for their impact. The project explores options with the vendor and alternatives, and any changes go through change control. Rushing to avoid the rules would be a compliance breach.

**Q17 — Answer: A** · *Process · PR5 Plan and manage procurement · Hybrid*

*Correct:* A. Assess the claim against the contract and the evidence, and negotiate a fair settlement

Claims administration assesses claims on the contract terms and the facts, and aims for a fair settlement, using dispute resolution only if needed.

**Q18 — Answer: B** · *People · PE4 Engage stakeholders · Predictive*

*Correct:* B. Tailor the methods and frequency of engagement to each group

Engagement is tailored to each stakeholder group's interests, influence and needs.

**Q19 — Answer: B** · *Process · PR2 Develop and manage project scope · Hybrid*

*Correct:* B. The product owner accepts the increments; the city formally accepts the milestones

In hybrid projects, increments are reviewed and accepted by the product owner, while contractual deliverables are formally accepted by the customer as defined in the contract.

**Q20 — Answer: B** · *People · PE2 Manage conflicts · Predictive*

*Correct:* B. Agree the fix now and settle responsibility separately under the contract

Separating the urgent technical fix from the commercial dispute keeps the project moving, while responsibility is resolved through the contract's mechanisms.

**Q21 — Answer: A, D** · *People · PE1 Develop a common vision · Agile*

*Correct:* A. It is short and memorable enough to guide everyday decisions / D. It describes the target customers and the value they will get

An effective vision is customer-focused, inspiring and brief, so that people can use it to make decisions. Feature lists and schedules belong elsewhere.

**Q22 — Answer: C** · *Business Environment · BE1 Define and establish project governance · Predictive*

*Correct:* C. Terms of reference: purpose, membership, decision rights and meeting rules

Terms of reference define how a governance body works and what it may decide, which makes its decisions faster and clearer.

**Q23 — Answer: A** · *Process · PR9 Evaluate project status · Hybrid*

*Correct:* A. Define objective thresholds for each colour on each measure

Objective thresholds make RAG status consistent and comparable, and reduce optimistic reporting.

**Q24 — Answer: A** · *Process · PR1 Develop an integrated project management plan and plan delivery · Hybrid*

*Correct:* A. Align the sprint outputs with the evidence needed for each stage gate

Hybrid planning connects the two approaches: iterative work is organised so that it produces the evidence required at each gate.

**Q25 — Answer: B** · *Business Environment · BE6 Continuous improvement · Agile*

*Correct:* B. Make explicit policies for each column, such as entry and exit criteria

Explicit policies make the rules of the workflow clear, which improves flow and makes improvement possible.

**Q26 — Answer: B** · *Business Environment · BE1 Define and establish project governance · Agile*

*Correct:* B. Consent-based decision-making

Consent-based decision-making moves forward unless there is a reasoned objection, which is faster than full consensus while respecting concerns.

**Q27 — Answer: C** · *Process · PR1 Develop an integrated project management plan and plan delivery · Agile*

*Correct:* C. To prove the game's quality, fun and production pipeline on a small but complete section

A vertical slice is a small, fully finished part of the product that validates the experience and the way the team will produce the rest.

**Q28 — Answer: B** · *Process · PR9 Evaluate project status · Predictive*

*Correct:* B. The project is falling further behind; the causes should be investigated and acted on

A widening gap between planned and actual progress is a negative trend that needs analysis and corrective action.

**Q29 — Answer: A** · *People · PE5 Align stakeholder expectations · Hybrid*

*Correct:* A. Attaching detailed specifications to the sales contracts and showing a model apartment

Clear written specifications and a physical example give buyers an accurate picture of what they will receive.

**Q30 — Answer: C** · *Process · PR6 Plan and manage finance · Predictive*

*Correct:* C. An S-curve

Spending usually starts slowly, peaks in the middle of the project and tails off at the end, which creates an S-shaped cumulative curve.

**Q31 — Answer: B** · *People · PE2 Manage conflicts · Predictive*

*Correct:* B. Interests rather than positions

Principled negotiation separates the people from the problem, focuses on interests rather than positions, generates options for mutual gain and uses objective criteria.

**Q32 — Answer: B** · *Process · PR1 Develop an integrated project management plan and plan delivery · Hybrid*

*Correct:* B. Pilot in one district with clear criteria, then scale based on results

When a solution is new to the organisation, a pilot with clear evaluation criteria reduces risk and informs the plan for the wider rollout.

**Q33 — Answer: B** · *People · PE4 Engage stakeholders · Predictive*

*Correct:* B. Early, with clear information, a contact person and practical relocation support

People strongly affected by a project need early, clear and supportive engagement. Late or purely legal engagement creates resistance and harm.

**Q34 — Answer: B** · *People · PE3 Lead the project team · Agile*

*Correct:* B. Facilitated the team in defining its own working agreements

Self-organising teams own their working agreements. Leaders facilitate and set boundaries, but people commit more to agreements they help create.

**Q35 — Answer: B** · *Process · PR9 Evaluate project status · Predictive*

*Correct:* B. Because SPI moves towards 1.0 as work is completed, even if the project is late

At completion, EV equals PV (both equal BAC), so SPI reaches 1.0 regardless of lateness. Other measures, such as earned schedule, give a better picture.

**Q36 — Answer: C** · *Process · PR5 Plan and manage procurement · Agile*

*Correct:* C. Tie them to accepted playable builds at agreed milestones

Paying for accepted, working increments aligns the vendor's incentives with delivering usable results.

**Q37 — Answer: C** · *Process · PR10 Manage project closure · Hybrid*

*Correct:* C. Cancelling project-specific contracts, licences and facilities

Closing administrative arrangements, such as leases, subscriptions and contracts, stops costs that the organisation no longer needs to pay.

**Q38 — Answer: D** · *Business Environment · BE5 Plan and manage risk · Hybrid*

*Correct:* D. Work with finance on responses, such as fixing the interest rate or hedging

Financial risks are handled with the right experts. Fixing rates or hedging transfers or reduces the exposure.

**Q39 — Answer: A** · *Business Environment · BE8 Evaluate external business environment changes · Agile*

*Correct:* A. Assess the impact on the business model and adapt the backlog and roadmap

Changes in the market environment are assessed for their effect on value, and the product strategy and backlog are adapted.

**Q40 — Answer: C** · *People · PE8 Plan and manage communication · Agile*

*Correct:* C. A visible dashboard showing the build status

An information radiator, such as a build status dashboard, makes problems visible immediately so that the team can react.

**Q41 — Answer: C** · *People · PE6 Manage stakeholder expectations · Agile*

*Correct:* C. Acknowledge the problem quickly and share regular, honest updates on the fixes

Fast acknowledgement and transparent updates manage expectations and preserve trust during incidents.

**Q42 — Answer: B** · *People · PE2 Manage conflicts · Hybrid*

*Correct:* B. Calm the discussion, acknowledge the concerns and refocus on the shared goal of road safety

Facilitating respectfully, acknowledging concerns and focusing on common interests turns a confrontation into a constructive discussion.

**Q43 — Answer: A** · *Process · PR7 Plan and optimize quality of products/deliverables · Hybrid*

*Correct:* A. Add field tests in realistic conditions to the acceptance criteria

Quality requirements must reflect real operating conditions. Testing in realistic conditions finds problems that laboratory tests miss.

**Q44 — Answer: D** · *People · PE3 Lead the project team · Predictive*

*Correct:* D. Maslow's hierarchy of needs

According to Maslow, unmet lower-level needs, such as security, take priority over higher-level needs, such as esteem and recognition.

**Q45 — Answer: D** · *Business Environment · BE2 Plan and manage project compliance · Agile*

*Correct:* D. Generate and maintain a software bill of materials as part of the build pipeline

Automating the software bill of materials keeps it accurate as the product changes, with little extra effort.

**Q46 — Answer: B** · *Process · PR6 Plan and manage finance · Predictive*

*Correct:* B. In the cost management plan

The cost management plan describes how costs will be planned, structured and controlled, including thresholds, units and precision.

**Q47 — Answer: C** · *People · PE4 Engage stakeholders · Predictive*

*Correct:* C. Through patient advocacy groups, for example in an advisory board

Patient advocacy groups and advisory boards bring the patients' perspective into design decisions, which can improve recruitment and relevance.

**Q48 — Answer: B** · *Business Environment · BE1 Define and establish project governance · Hybrid*

*Correct:* B. A design authority that approves technical decisions affecting more than one vendor

A design authority provides technical governance across parties, ensuring consistent decisions on shared architecture and interfaces.

**Q49 — Answer: C** · *Process · PR8 Plan and manage schedule · Agile*

*Correct:* C. Affinity estimating, grouping items by relative size

Affinity estimating lets a team sort many items into relative size groups quickly, which is useful for large backlogs.

**Q50 — Answer: C** · *People · PE2 Manage conflicts · Agile*

*Correct:* C. Facilitate using data and team norms, and take scope options to the product owner

Using facts and agreed norms keeps the discussion on the problem. The real trade-off, scope versus date, is a product decision.

**Q51 — Answer: B** · *People · PE2 Manage conflicts · Predictive*

*Correct:* B. Bring both together to agree a sequenced work plan for each floor

Coordinating the sequence of work with both parties removes the cause of the conflict and improves flow.

**Q52 — Answer: A** · *Process · PR4 Plan and manage resources · Predictive*

*Correct:* A. Resequence the trades so that fewer work on each floor at a time

Planning the flow of trades through the building, for example with takt planning, limits congestion and improves safety and productivity.

**Q53 — Answer: A, D** · *People · PE8 Plan and manage communication · Hybrid*

*Correct:* A. An open data portal / D. A project intranet site

With pull communication, stakeholders access information when they need it. Emails are push communication; calls and meetings are interactive.

**Q54 — Answer: B** · *People · PE7 Help ensure knowledge transfer · Agile*

*Correct:* B. Short "three amigos" conversations about each story

In three amigos sessions, business, development and testing perspectives discuss each story together, which surfaces misunderstandings before work begins.

**Q55 — Answer: D, E** · *Business Environment · BE5 Plan and manage risk · Hybrid*

*Correct:* D. It is specific enough to be analysed and responded to / E. It describes the cause, the uncertain event and its effect

A clear "cause–risk–effect" statement makes it possible to assess the risk and choose a targeted response.

**Q56 — Answer: A** · *People · PE6 Manage stakeholder expectations · Agile*

*Correct:* A. Show the value of the technical work, for example through performance results

Technical work also delivers value. Making it visible, for example through improved speed or stability, keeps stakeholders informed and supportive.

**Q57 — Answer: A** · *People · PE3 Lead the project team · Predictive*

*Correct:* A. Define the outcome, the authority, the resources and the check-in points

Effective delegation makes the expected outcome, the authority and the support clear, with agreed check-ins rather than micromanagement.

**Q58 — Answer: C** · *People · PE8 Plan and manage communication · Agile*

*Correct:* C. Agree communication norms, such as tone and when to switch to a call

Team norms for written communication, including when to talk instead, reduce misunderstandings in remote teams.

**Q59 — Answer: D** · *Process · PR8 Plan and manage schedule · Predictive*

*Correct:* D. At least six weeks, allowing for one resubmission

Planning must include the typical resubmission: 4 weeks of certification plus 2 weeks for one resubmission means submitting at least 6 weeks before launch.

**Q60 — Answer: D** · *Process · PR7 Plan and optimize quality of products/deliverables · Agile*

*Correct:* D. Fix the five most frequent crash types, tracking the crash rate on every build

Most crashes come from a few causes, so fixing them gives the largest improvement. Tracking the rate on every build shows whether the target is being reached.

**Q61 — Answer: D** · *Process · PR5 Plan and manage procurement · Hybrid*

*Correct:* D. Reject the non-conforming models under the contract specification and address the cause with the vendor

The contract's specification is the basis for acceptance. Rejecting non-conforming work and addressing the cause with the vendor protects quality and the contract.

**Q62 — Answer: C** · *Process · PR3 Help ensure value-based delivery · Agile*

*Correct:* C. Player retention over time, such as the share still playing after 30 days

For a live service, continued engagement shows lasting value. First-week sales and output measures do not show whether players stay.

**Q63 — Answer: D** · *People · PE7 Help ensure knowledge transfer · Hybrid*

*Correct:* D. A structured handover to her successor, including introductions to key stakeholders

Relationship knowledge is tacit. Personal introductions and a structured handover transfer it far better than lists or documents.

**Q64 — Answer: C** · *Process · PR3 Help ensure value-based delivery · Hybrid*

*Correct:* C. Install the LEDs first, so that the energy savings start early

Sequencing work so that value is delivered early starts benefits sooner and reduces risk.

**Q65 — Answer: D** · *Business Environment · BE3 Manage and control changes · Predictive*

*Correct:* D. The sponsor, or the body that issued it

The charter is issued by the sponsor, so changes to it are approved by the sponsor or the governance body that authorised the project.

**Q66 — Answer: D** · *Process · PR2 Develop and manage project scope · Agile*

*Correct:* D. The sprint goal, the selected items and a plan for delivering them

The sprint backlog is the developers' plan for the sprint: the sprint goal, the product backlog items selected and how they will be delivered.

**Q67 — Answer: 1→B, 2→D, 3→A, 4→C** · *People · PE3 Lead the project team · Predictive*

*Correct pairs:* 1. Transformational → Inspires with a vision and encourages innovation; 2. Transactional → Uses rewards and corrective action based on performance; 3. Servant → Puts the team's needs first and removes obstacles; 4. Laissez-faire → Leaves decisions largely to the team, with little involvement

Effective project leaders adapt their style to the situation and the team's needs.

**Q68 — Answer: C** · *Process · PR3 Help ensure value-based delivery · Agile*

*Correct:* C. Interview a sample of property managers about the problem

Product discovery starts by validating the problem with real users before investing in a solution.

**Q69 — Answer: A** · *Process · PR4 Plan and manage resources · Agile*

*Correct:* A. A small platform team that provides shared tools to the other teams

A platform team can provide shared tools and services as a product for other teams, reducing duplication and cognitive load.

**Q70 — Answer: C** · *People · PE3 Lead the project team · Hybrid*

*Correct:* C. Consultative: gather the team's input, then decide

A consultative style combines speed with the team's expertise: the leader seeks input but remains responsible for the decision.

**Q71 — Answer: D** · *People · PE8 Plan and manage communication · Predictive*

*Correct:* D. Use translated materials, visual signs and interpreters, and check understanding

Safety communication must be understood by everyone. Translations, visuals, interpreters and checks for understanding make that possible.

**Q72 — Answer: C** · *Process · PR6 Plan and manage finance · Predictive*

*Correct:* C. An allowance for expected price increases over the period

Escalation (inflation) allowances account for expected price changes over long projects.

**Q73 — Answer: B** · *People · PE6 Manage stakeholder expectations · Predictive*

*Correct:* B. Revisit the plan and communications with the investor to realign them

Stakeholder expectations change over time. Monitoring them and realigning plans and communications keeps the project relevant.

**Q74 — Answer: C** · *Process · PR10 Manage project closure · Predictive*

*Correct:* C. An evaluation of the vendor's performance

Vendor performance evaluations become organisational process assets that inform future source selection.

**Q75 — Answer: A** · *Process · PR3 Help ensure value-based delivery · Predictive*

*Correct:* A. Occupancy and rental income compared with the business case

The investor's value comes from the building's use: occupancy and rental income show whether the business case is being achieved.

**Q76 — Answer: B** · *Business Environment · BE5 Plan and manage risk · Agile*

*Correct:* B. Assess the update early in a separate branch and plan when to adopt it

Early, low-risk investigation (a spike in a separate branch) reveals the update's impact, so that adoption can be planned.

**Q77 — Answer: C** · *Process · PR7 Plan and optimize quality of products/deliverables · Predictive*

*Correct:* C. About 99.7%

For a normal distribution, about 68.3% of values fall within ±1 sigma, 95.4% within ±2 sigma and 99.7% within ±3 sigma.

**Q78 — Answer: C** · *Process · PR9 Evaluate project status · Agile*

*Correct:* C. 90%

Defect removal efficiency = defects found before release / total defects = 90 / (90 + 10) = 90%.

**Q79 — Answer: C** · *Business Environment · BE6 Continuous improvement · Hybrid*

*Correct:* C. CAPA (corrective and preventive action)

A CAPA system investigates problems, corrects them and prevents recurrence, and is a core continuous improvement mechanism in regulated industries.

**Q80 — Answer: D** · *People · PE2 Manage conflicts · Agile*

*Correct:* D. Coach the lead designer and use inclusive techniques, such as silent brainstorming

Coaching and inclusive facilitation techniques give everyone a voice and address the behaviour constructively.

**Q81 — Answer: B** · *Business Environment · BE1 Define and establish project governance · Hybrid*

*Correct:* B. An independent data monitoring committee

An independent data monitoring committee protects patients by reviewing safety data and recommending whether the trial should continue, change or stop.

**Q82 — Answer: A** · *Business Environment · BE6 Continuous improvement · Predictive*

*Correct:* A. About 3.4

Six sigma performance corresponds to about 3.4 defects per million opportunities, allowing for a long-term shift in the process mean.

**Q83 — Answer: C** · *People · PE1 Develop a common vision · Agile*

*Correct:* C. Facilitate agreement on a few design pillars and success measures that balance both views

Shared design pillars and agreed success measures give everyone the same reference for decisions. Letting one side win or splitting the work does not create a common vision.

**Q84 — Answer: D** · *Business Environment · BE7 Support organizational change · Predictive*

*Correct:* D. Explain the reasons, involve employees in designing team zones, and pilot the layout

Involvement, explanation and piloting address the reasons for resistance and allow the new way of working to be adapted to people's needs.

**Q85 — Answer: D** · *Process · PR4 Plan and manage resources · Predictive*

*Correct:* D. Exactly one

Each deliverable has exactly one accountable person, which avoids confusion about who has the final say.

**Q86 — Answer: D** · *Process · PR2 Develop and manage project scope · Agile*

*Correct:* D. Add enabler work to externalise the text, prioritised, and treat localisation as a requirement

Missed requirements are added to the backlog and prioritised. Enabler work makes the product ready for localisation.

**Q87 — Answer: B** · *Business Environment · BE7 Support organizational change · Predictive*

*Correct:* B. Plan and hold the consultation before implementing the change

Legal consultation requirements are planned into the change, with enough time for genuine consultation before implementation.

**Q88 — Answer: D** · *Process · PR4 Plan and manage resources · Hybrid*

*Correct:* D. Plan flexible capacity, such as an external test partner, for the peak

Short peaks in demand are best handled with flexible resources, planned in advance, rather than permanent hires or reduced quality.

**Q89 — Answer: D** · *People · PE2 Manage conflicts · Hybrid*

*Correct:* D. Agree the analysis method together, and involve an independent statistician if needed

Disagreements about data are best resolved by agreeing objective methods and, if necessary, independent expertise.

**Q90 — Answer: B** · *Process · PR10 Manage project closure · Predictive*

*Correct:* B. An occupancy certificate from the authorities

Buildings normally need an occupancy certificate, issued after inspections, before they can legally be used.

**Q91 — Answer: A** · *Process · PR6 Plan and manage finance · Hybrid*

*Correct:* A. A quantity variance: the team should investigate why more concrete was needed

Separating price and quantity effects shows the true cause. Here, the extra usage needs investigation, for example design changes or waste.

**Q92 — Answer: C** · *Process · PR5 Plan and manage procurement · Predictive*

*Correct:* C. Design-build

In design-build, one contract covers both design and construction, which gives a single point of responsibility.

**Q93 — Answer: A, B** · *People · PE3 Lead the project team · Agile*

*Correct:* A. Making overtime visible and limiting it / B. Planning work based on sustainable capacity

Sustainable pace depends on realistic planning and on treating overtime as an exception that is visible and limited.

**Q94 — Answer: B** · *Process · PR8 Plan and manage schedule · Agile*

*Correct:* B. Splitting the items into smaller pieces

Smaller items flow through the process faster and more predictably, which reduces cycle time.

**Q95 — Answer: A** · *Process · PR1 Develop an integrated project management plan and plan delivery · Agile*

*Correct:* A. The developers, at the daily scrum

The developers plan their own daily work to progress towards the sprint goal, typically at the daily scrum.

**Q96 — Answer: B, D** · *People · PE4 Engage stakeholders · Hybrid*

*Correct:* B. Reviewing the registers of similar past projects / D. Asking known stakeholders who else is affected

Snowball interviews and past project records reveal stakeholders who might otherwise be missed.

**Q97 — Answer: A** · *Business Environment · BE2 Plan and manage project compliance · Hybrid*

*Correct:* A. Report it as a GCP deviation, assess the impact and retrain the site staff

Informed consent is a fundamental requirement. Deviations are reported and investigated, and corrective actions are taken. Back-dating records is falsification.

**Q98 — Answer: B** · *People · PE7 Help ensure knowledge transfer · Predictive*

*Correct:* B. A structured technology transfer with documents, training, joint runs and checks

Technology transfer combines explicit knowledge (documents) with tacit knowledge (training and joint runs) and verifies that the receiver can perform the process.

**Q99 — Answer: D** · *People · PE3 Lead the project team · Agile*

*Correct:* D. Help the team raise it directly with the colleague, referring to their working agreements

Self-organising teams address performance issues through direct, respectful feedback based on their own agreements. Escalation follows only if this does not work.

**Q100 — Answer: C** · *People · PE1 Develop a common vision · Predictive*

*Correct:* C. Revisit the charter with the key stakeholders in a workshop

Developing the charter collaboratively builds shared understanding of the objectives and commitment to them.

**Q101 — Answer: D** · *Process · PR9 Evaluate project status · Agile*

*Correct:* D. More work is being started in parallel, or stories are waiting longer between steps

Rising cycle time with stable throughput suggests higher work in progress or more waiting. The team should examine its flow and WIP.

**Q102 — Answer: D** · *People · PE7 Help ensure knowledge transfer · Predictive*

*Correct:* D. As-built drawings, operation and maintenance manuals, and warranties

Facility managers need accurate as-built information, operating and maintenance instructions and warranty details to run the building.

**Q103 — Answer: $840,000** · *Process · PR6 Plan and manage finance · Predictive*

EV = percent complete × BAC = 0.35 × 2,400,000 = $840,000.

**Q104 — Answer: B** · *Business Environment · BE8 Evaluate external business environment changes · Hybrid*

*Correct:* B. Reassess the business case and the building's mix of uses with the sponsor

Changes in the market can change the best use of an asset. The business case is reassessed so that the sponsor can decide on the best option.

**Q105 — Answer: A** · *Process · PR10 Manage project closure · Hybrid*

*Correct:* A. Notify the authorities, close the sites, archive the data and record lessons

Early termination still requires a formal close-out that follows regulatory requirements and preserves the data and lessons.

**Q106 — Answer: D** · *Process · PR7 Plan and optimize quality of products/deliverables · Predictive*

*Correct:* D. A snagging (punch) list

A snagging or punch list records the outstanding defects and incomplete items that must be fixed before or soon after handover.

**Q107 — Answer: A** · *Business Environment · BE4 Remove impediments and manage issues · Agile*

*Correct:* A. Raise it as a cross-team impediment and agree environment rules or separate environments

Shared resources need agreed rules or redesign. The scrum master works across teams to remove the impediment at its source.

**Q108 — Answer: A** · *People · PE8 Plan and manage communication · Predictive*

*Correct:* A. Formally, through documented submissions and meetings, via regulatory affairs

Regulatory communication is formal, documented and coordinated, usually by a regulatory affairs function.

**Q109 — Answer: D** · *People · PE3 Lead the project team · Hybrid*

*Correct:* D. GROW

The GROW model (goal, reality, options, will or way forward) structures coaching conversations so that the person finds their own solution.

**Q110 — Answer: C** · *People · PE6 Manage stakeholder expectations · Predictive*

*Correct:* C. Inform the buyers now, explain why and help with options, such as letters for lenders

Early, honest communication with practical support lets stakeholders manage the impact of a delay on them.

**Q111 — Answer: B** · *Process · PR8 Plan and manage schedule · Predictive*

*Correct:* B. Finish no later than

A "finish no later than" constraint sets the latest date by which an activity must be complete. Finish-to-start is a dependency type, not a constraint.

**Q112 — Answer: D** · *Process · PR10 Manage project closure · Predictive*

*Correct:* D. The final account, including all variations and claims

The final account settles the total amount due under the contract, including approved variations and resolved claims.

**Q113 — Answer: D** · *People · PE5 Align stakeholder expectations · Agile*

*Correct:* D. Agree honest messaging: show only features likely to ship, or label them as in progress

Honest communication sets realistic expectations and protects the studio's reputation. Promising features that may be cut leads to disappointment.

**Q114 — Answer: B** · *People · PE4 Engage stakeholders · Hybrid*

*Correct:* B. Engage them early, share the privacy design and invite their review

Engaging critics transparently can improve the design and build trust. Ignoring or threatening them increases opposition.

**Q115 — Answer: A** · *Business Environment · BE4 Remove impediments and manage issues · Predictive*

*Correct:* A. Review and prioritise the issues regularly with their owners, closing or escalating them

An issue log is only useful when it is actively managed: issues are prioritised, owned, resolved, escalated or closed.

**Q116 — Answer: A** · *Business Environment · BE2 Plan and manage project compliance · Predictive*

*Correct:* A. Stop work in the area and follow the legal requirements for removal by licensed specialists

Hazardous materials must be handled according to the law, by licensed specialists. Safety and compliance come before the schedule.

**Q117 — Answer: D** · *People · PE5 Align stakeholder expectations · Predictive*

*Correct:* D. Clarify the specification with the buyers and offer upgrades as paid changes

Expectations are aligned with the agreed specification. Additional requests are handled as changes, with their cost and timing made clear.

**Q118 — Answer: A** · *Process · PR5 Plan and manage procurement · Predictive*

*Correct:* A. Unit price

A unit price contract sets a fixed rate per unit of work, which suits work with well-defined units but uncertain quantities.

**Q119 — Answer: A** · *Process · PR7 Plan and optimize quality of products/deliverables · Agile*

*Correct:* A. Static code analysis

Static analysis examines source code without executing it, finding common defects and standard violations early in the pipeline.

**Q120 — Answer: B** · *Business Environment · BE3 Manage and control changes · Hybrid*

*Correct:* B. A protocol amendment is approved by the regulator and the ethics committee

In regulated trials, substantial protocol changes require formal amendments approved by the regulator and the ethics committee before implementation.

**Q121 — Answer: 18** · *Process · PR8 Plan and manage schedule · Predictive*

Path A-B-E = 5 + 7 + 3 = 15 days. Path A-C-D-E = 5 + 4 + 6 + 3 = 18 days. The critical path is A-C-D-E, so the project takes 18 days.

**Q122 — Answer: A** · *People · PE1 Develop a common vision · Hybrid*

*Correct:* A. Create shared goals and rituals around the game, such as joint demos and a common team charter

Shared goals, rituals and agreements build a common identity across locations. Ranking studios against each other would deepen the divisions.

**Q123 — Answer: A** · *Process · PR4 Plan and manage resources · Hybrid*

*Correct:* A. They are trained on the study's procedures and their qualifications are confirmed

Regulated work requires qualified, trained staff. Training and qualification checks are part of acquiring and developing the team.

**Q124 — Answer: 40** · *Process · PR3 Help ensure value-based delivery · Predictive*

Risk-adjusted value = 0.20 × 500 − 60 = 100 − 60 = 40 (million dollars).

**Q125 — Answer: B** · *Process · PR3 Help ensure value-based delivery · Agile*

*Correct:* B. Those with the lowest value relative to their effort

Value-based prioritisation removes the items that contribute least value for the effort they require.

**Q126 — Answer: C** · *People · PE7 Help ensure knowledge transfer · Agile*

*Correct:* C. Regular internal tech talks, recorded for later viewing

Internal talks share specialist knowledge widely and cheaply, and recordings make the knowledge available to people who join later.

**Q127 — Answer: A** · *Process · PR8 Plan and manage schedule · Hybrid*

*Correct:* A. Takt time

Takt time is the pace of production needed to meet demand or a deadline. Trades move through the building at this rhythm.

**Q128 — Answer: D** · *Business Environment · BE4 Remove impediments and manage issues · Predictive*

*Correct:* D. Escalate with the power company and plan workarounds, such as generators

External issues are escalated through the right channels, while the project limits the impact with safe, approved workarounds.

**Q129 — Answer: B** · *Business Environment · BE3 Manage and control changes · Hybrid*

*Correct:* B. Report the cumulative impact and review the remaining reserve and priorities with the sponsor

Many small changes can have a large combined effect. Making the cumulative impact visible allows governance to decide how to proceed.

**Q130 — Answer: B, C** · *Process · PR3 Help ensure value-based delivery · Hybrid*

*Correct:* B. Vehicle emissions compared with a baseline / C. Average journey times compared with a baseline

Value is shown by outcomes, such as shorter journeys and lower emissions, measured against a baseline. Installation counts and spending are outputs and inputs.

**Q131 — Answer: A** · *Business Environment · BE3 Manage and control changes · Predictive*

*Correct:* A. A full impact assessment, including the impact on data protection

Impact analysis covers all affected areas, including legal and privacy impacts, so that the board can make an informed decision.

**Q132 — Answer: D** · *Business Environment · BE4 Remove impediments and manage issues · Hybrid*

*Correct:* D. Work with the sites to remove the barriers, such as complex forms

Understanding why the issue occurs and removing its causes is more effective than pressure or replacement.

**Q133 — Answer: B** · *People · PE4 Engage stakeholders · Agile*

*Correct:* B. Through community channels and early-access builds, with feedback feeding the backlog

Early and continuous engagement with players provides feedback when it is still cheap to act on it.

**Q134 — Answer: A** · *People · PE6 Manage stakeholder expectations · Hybrid*

*Correct:* A. Inform the communications team now and agree revised messaging and options, such as a preview

Stakeholders who have made commitments based on the plan need early warning, so that they can adjust messages and plans together with the project.

**Q135 — Answer: C, E** · *Process · PR1 Develop an integrated project management plan and plan delivery · Predictive*

*Correct:* C. The schedule baseline / E. The cost baseline

The scope, schedule and cost baselines are approved versions used to measure performance. Registers and logs are project documents that change frequently.

**Q136 — Answer: C** · *People · PE5 Align stakeholder expectations · Predictive*

*Correct:* C. Use the contract's definition and the agreed evidence, such as an occupancy permit

Contract definitions and agreed evidence provide the objective basis for aligning expectations about completion.

**Q137 — Answer: D** · *People · PE1 Develop a common vision · Hybrid*

*Correct:* D. Review with the team how this project differs, using the data, and reaffirm the vision

A shared vision needs to survive new information. An honest, evidence-based discussion of the differences rebuilds confidence, or shows if the direction should change.

**Q138 — Answer: B** · *Process · PR6 Plan and manage finance · Agile*

*Correct:* B. Monitor the cost per player and optimise the architecture, with alerts on spending

Operating costs that scale with usage are controlled through unit-cost metrics, technical optimisation and spending alerts.

**Q139 — Answer: C** · *Process · PR7 Plan and optimize quality of products/deliverables · Agile*

*Correct:* C. Playtesting with real players, observing them as they play

Playtesting is the game industry's form of usability testing: observing real players shows whether the experience works as intended.

**Q140 — Answer: D** · *Business Environment · BE8 Evaluate external business environment changes · Hybrid*

*Correct:* D. Assess whether the drug is eligible and how the pathway would change the plan

External changes can create opportunities as well as threats. They are assessed and, if beneficial, built into the strategy and plan.

**Q141 — Answer: B** · *Process · PR2 Develop and manage project scope · Predictive*

*Correct:* B. A user requirements specification

A user requirements specification defines what the instrument must do, which is the basis for selection, qualification and acceptance.

**Q142 — Answer: C** · *Business Environment · BE1 Define and establish project governance · Predictive*

*Correct:* C. Build them into the governance framework and the plan, with clear owners

Funding conditions are obligations. Building them into governance, plans and responsibilities ensures that they are met and the funding is not put at risk.

**Q143 — Answer: C** · *Process · PR9 Evaluate project status · Predictive*

*Correct:* C. 14 months

Remaining patients = 120 − 36 = 84. At 6 per month, 84 / 6 = 14 more months.

**Q144 — Answer: A** · *Process · PR8 Plan and manage schedule · Predictive*

*Correct:* A. Merge bias

Merge bias means that the more paths converge at a point, the lower the probability that all of them finish on time.

**Q145 — Answer: D** · *Business Environment · BE1 Define and establish project governance · Agile*

*Correct:* D. Agree outcome-based milestone criteria, such as playable builds, and plan freely in between

Outcome-based milestones give the funder control over investment decisions while leaving the team free to decide how to reach them.

**Q146 — Answer: D** · *Business Environment · BE6 Continuous improvement · Agile*

*Correct:* D. A regular team health check

Team health checks give teams a quick, regular self-assessment that highlights trends and areas for improvement.

**Q147 — Answer: C** · *People · PE6 Manage stakeholder expectations · Predictive*

*Correct:* C. Explain the freeze and offer options, with cost and timing, through the change process

Expectations are managed by explaining the constraints and offering a clear, fair process for any changes that remain possible.

**Q148 — Answer: B** · *Business Environment · BE5 Plan and manage risk · Agile*

*Correct:* B. A feature toggle

Feature toggles separate deployment from release, so that a feature can be turned off quickly without a new deployment.

**Q149 — Answer: B** · *Process · PR9 Evaluate project status · Hybrid*

*Correct:* B. Raise a change request with the analysis and options

Performance analysis leads to recommended corrective actions. Actions that affect the baselines are submitted as change requests.

**Q150 — Answer: B** · *Process · PR6 Plan and manage finance · Hybrid*

*Correct:* B. $750,000

Additional cost = (150 − 120) × 25,000 = 30 × 25,000 = $750,000.

**Q151 — Answer: B** · *Process · PR1 Develop an integrated project management plan and plan delivery · Predictive*

*Correct:* B. A project management information system

A project management information system provides tools for scheduling, document management, reporting and performance tracking.

**Q152 — Answer: A** · *People · PE7 Help ensure knowledge transfer · Hybrid*

*Correct:* A. Through site visits and peer exchange before finalising the design

Learning from peers who have done similar work is a valuable source of lessons, especially when it is gathered early.

**Q153 — Answer: C** · *People · PE4 Engage stakeholders · Hybrid*

*Correct:* C. With short, outcome-focused briefings at key decision points

Busy, powerful stakeholders need concise information focused on outcomes and decisions, timed for when their input matters.

**Q154 — Answer: D** · *Business Environment · BE3 Manage and control changes · Agile*

*Correct:* D. Show the impact on the release forecast and agree trade-offs with stakeholders

Scope changes in agile are managed through the backlog. Making the impact visible lets stakeholders make informed trade-offs.

**Q155 — Answer: A** · *Business Environment · BE7 Support organizational change · Agile*

*Correct:* A. Provide training and time to learn, with senior artists as mentors

Change temporarily reduces productivity. Training, mentoring and realistic expectations help people through the transition.

**Q156 — Answer: A** · *Process · PR2 Develop and manage project scope · Hybrid*

*Correct:* A. A facilitated requirements workshop

Facilitated workshops bring stakeholders together to define requirements, resolve conflicts and build consensus quickly.

**Q157 — Answer: D** · *People · PE3 Lead the project team · Agile*

*Correct:* D. Peer-to-peer recognition, for example thanking colleagues in retrospectives

Peer recognition reinforces collaboration and is timely. Rewards for individual output can undermine teamwork.

**Q158 — Answer: 4** · *Business Environment · BE5 Plan and manage risk · Predictive*

Expected delay = 0.25 × 8 + 0.50 × 4 = 2 + 2 = 4 days.

**Q159 — Answer: D** · *People · PE5 Align stakeholder expectations · Predictive*

*Correct:* D. Explain the regulatory steps and timelines, and agree realistic milestones

Expectations about time must reflect external realities such as regulatory processes. Honest milestone planning keeps trust.

**Q160 — Answer: C** · *Process · PR6 Plan and manage finance · Agile*

*Correct:* C. Analogous estimating based on relative size

Using historical costs of similar-sized items is a form of analogous, relative estimating that is fast and suits early decisions.

**Q161 — Answer: C** · *Process · PR2 Develop and manage project scope · Predictive*

*Correct:* C. In the scope statement and the tenants' lease specification

Scope boundaries, including what is excluded, are defined in the scope statement and the contracts, so that all parties share the same understanding.

**Q162 — Answer: C** · *Business Environment · BE7 Support organizational change · Hybrid*

*Correct:* C. Raise it with the sponsor and the leaders to align their messages and behaviour

Mixed messages from leaders undermine change. Aligning leaders' words and actions, with the sponsor's support, is essential.

**Q163 — Answer: A** · *Business Environment · BE6 Continuous improvement · Hybrid*

*Correct:* A. Reference class forecasting

Reference class forecasting bases estimates on the actual results of a class of similar projects, which corrects optimism bias.

**Q164 — Answer: D** · *Process · PR10 Manage project closure · Agile*

*Correct:* D. Confirming the licence terms and transferring the licences for live operations

Closure includes administrative transfers. Licences needed to operate the product must be confirmed and handed over to the operating team.

**Q165 — Answer: D** · *Business Environment · BE4 Remove impediments and manage issues · Agile*

*Correct:* D. Escalate the need for more kits, and agree a fair booking system in the meantime

The scrum master pursues a lasting fix for the impediment while reducing its impact through a fair, transparent arrangement.

**Q166 — Answer: A** · *Process · PR2 Develop and manage project scope · Predictive*

*Correct:* A. The 100% rule

The 100% rule says that the WBS includes all of the work defined by the project scope, and nothing outside it.

**Q167 — Answer: A** · *Process · PR5 Plan and manage procurement · Predictive*

*Correct:* A. Non-disclosure agreements

Non-disclosure agreements protect confidential information shared during the selection process, before any contract is signed.

**Q168 — Answer: 1→B, 2→A, 3→C, 4→D** · *Business Environment · BE5 Plan and manage risk · Predictive*

*Correct pairs:* 1. Escalate → Pass the threat to a higher level that owns it; 2. Avoid → Change the plan to eliminate the threat; 3. Transfer → Shift the impact and ownership to a third party; 4. Mitigate → Reduce the probability or the impact of the threat

Accept is the fifth threat response: acknowledging the threat without action, possibly with a contingency reserve.

**Q169 — Answer: C** · *Process · PR1 Develop an integrated project management plan and plan delivery · Agile*

*Correct:* C. Must-have scope within a cautious forecast, with stretch items marked

With a fixed date, the release plan protects the essential scope by planning it within a cautious forecast, while lower-priority items are shown as possible extras.

**Q170 — Answer: A** · *Process · PR3 Help ensure value-based delivery · Predictive*

*Correct:* A. The discount rate at which the project's net present value equals zero

IRR is the discount rate that makes the NPV of the project's cash flows zero. A higher IRR generally indicates a more attractive investment.

**Q171 — Answer: C, D** · *People · PE2 Manage conflicts · Predictive*

*Correct:* C. Withdrawing (avoiding) / D. Smoothing (accommodating)

Smoothing plays down differences and withdrawing postpones the issue; neither resolves its cause. Collaborating addresses the root of the conflict.

**Q172 — Answer: A** · *Process · PR5 Plan and manage procurement · Hybrid*

*Correct:* A. Use a procedure that allows dialogue with bidders, within the rules

Public procurement rules often include procedures, such as competitive dialogue or innovation partnerships, for needs that cannot be fully specified in advance.

**Q173 — Answer: A** · *People · PE5 Align stakeholder expectations · Hybrid*

*Correct:* A. Clarify the update interval early and agree the requirement based on their real needs

Expectation gaps are surfaced early and resolved by understanding the real need and agreeing realistic requirements.

**Q174 — Answer: A, D** · *Business Environment · BE7 Support organizational change · Agile*

*Correct:* A. Making short-term wins visible and celebrating them / D. Using early successes to drive further improvements

Visible short-term wins maintain energy, and consolidating gains to produce more change prevents a return to old habits (Kotter's steps 6 and 7).

**Q175 — Answer: A** · *Process · PR4 Plan and manage resources · Predictive*

*Correct:* A. To speed up communication and decisions during a critical period

Co-location, sometimes in a "war room", improves communication and speeds up problem solving when time is critical.

**Q176 — Answer: D** · *People · PE1 Develop a common vision · Predictive*

*Correct:* D. A structured induction covering the project's objectives, values and safety rules

A consistent induction gives every newcomer the same understanding of the project's goals, expectations and rules from their first day.

**Q177 — Answer: C** · *Business Environment · BE2 Plan and manage project compliance · Agile*

*Correct:* C. Check the rules in each target market with legal experts and adapt the features per market

Products sold in several markets must comply with each market's rules. Legal input guides how features are adapted.

**Q178 — Answer: D** · *Process · PR4 Plan and manage resources · Agile*

*Correct:* D. Swarming

Swarming focuses the team on finishing the most important work first, which reduces work in progress and cycle time.

**Q179 — Answer: A** · *People · PE1 Develop a common vision · Agile*

*Correct:* A. To align the team on the customer outcome, working backwards from it

"Working backwards" from a future press release makes the customer benefit concrete and gives the team a shared picture of success.

**Q180 — Answer: B** · *People · PE8 Plan and manage communication · Predictive*

*Correct:* B. Send an agenda in advance, timebox the items and record actions with owners and dates

Good meeting management, with an agenda, timeboxes and recorded actions, makes meetings shorter and more effective.

## Score Analysis

Count your correct answers in each domain and task, then compute the percentage. A multiple-response or matching question counts as correct only if **every** part is right (as on the real exam).

| Domain | Questions | Your correct | % |
|---|---|---|---|
| People | 59 | | |
| Process | 74 | | |
| Business Environment | 47 | | |
| **Total** | **180** | | |

| Task | Questions (numbers) |
|---|---|
| PE1 — Develop a common vision | 7 (21, 83, 100, 122, 137, 176, 179) |
| PE2 — Manage conflicts | 8 (20, 31, 42, 50, 51, 80, 89, 171) |
| PE3 — Lead the project team | 9 (34, 44, 57, 67, 70, 93, 99, 109, 157) |
| PE4 — Engage stakeholders | 8 (4, 18, 33, 47, 96, 114, 133, 153) |
| PE5 — Align stakeholder expectations | 7 (3, 29, 113, 117, 136, 159, 173) |
| PE6 — Manage stakeholder expectations | 7 (2, 41, 56, 73, 110, 134, 147) |
| PE7 — Help ensure knowledge transfer | 6 (54, 63, 98, 102, 126, 152) |
| PE8 — Plan and manage communication | 7 (10, 40, 53, 58, 71, 108, 180) |
| PR1 — Develop an integrated project management plan and plan delivery | 8 (1, 24, 27, 32, 95, 135, 151, 169) |
| PR2 — Develop and manage project scope | 8 (7, 19, 66, 86, 141, 156, 161, 166) |
| PR3 — Help ensure value-based delivery | 8 (62, 64, 68, 75, 124, 125, 130, 170) |
| PR4 — Plan and manage resources | 7 (52, 69, 85, 88, 123, 175, 178) |
| PR5 — Plan and manage procurement | 7 (17, 36, 61, 92, 118, 167, 172) |
| PR6 — Plan and manage finance | 8 (30, 46, 72, 91, 103, 138, 150, 160) |
| PR7 — Plan and optimize quality of products/deliverables | 7 (11, 43, 60, 77, 106, 119, 139) |
| PR8 — Plan and manage schedule | 8 (8, 49, 59, 94, 111, 121, 127, 144) |
| PR9 — Evaluate project status | 7 (23, 28, 35, 78, 101, 143, 149) |
| PR10 — Manage project closure | 6 (37, 74, 90, 105, 112, 164) |
| BE1 — Define and establish project governance | 6 (22, 26, 48, 81, 142, 145) |
| BE2 — Plan and manage project compliance | 6 (5, 13, 45, 97, 116, 177) |
| BE3 — Manage and control changes | 6 (9, 65, 120, 129, 131, 154) |
| BE4 — Remove impediments and manage issues | 6 (12, 107, 115, 128, 132, 165) |
| BE5 — Plan and manage risk | 7 (14, 38, 55, 76, 148, 158, 168) |

| Task | Questions (numbers) |
|---|---|
| BE6 — Continuous improvement | 5 (25, 79, 82, 146, 163) |
| BE7 — Support organizational change | 6 (15, 84, 87, 155, 162, 174) |
| BE8 — Evaluate external business environment changes | 5 (6, 16, 39, 104, 140) |

**How to read your result** (self-assessment guide — PMI does not publish a fixed pass mark; the real exam reports *Above Target / Target / Below Target / Needs Improvement* per domain):

| Your % | Indicative rating | What to do |
|---|---|---|
| 80% or more | Above Target | You are ready. Keep practising timing. |
| 70–79% | Target | Likely pass. Review every missed question and its domain. |
| 60–69% | Below Target | Re-study the weak domains in the Study Guide before the next exam. |
| under 60% | Needs Improvement | Go back to the Study Guide chapters and the Question Bank first. |